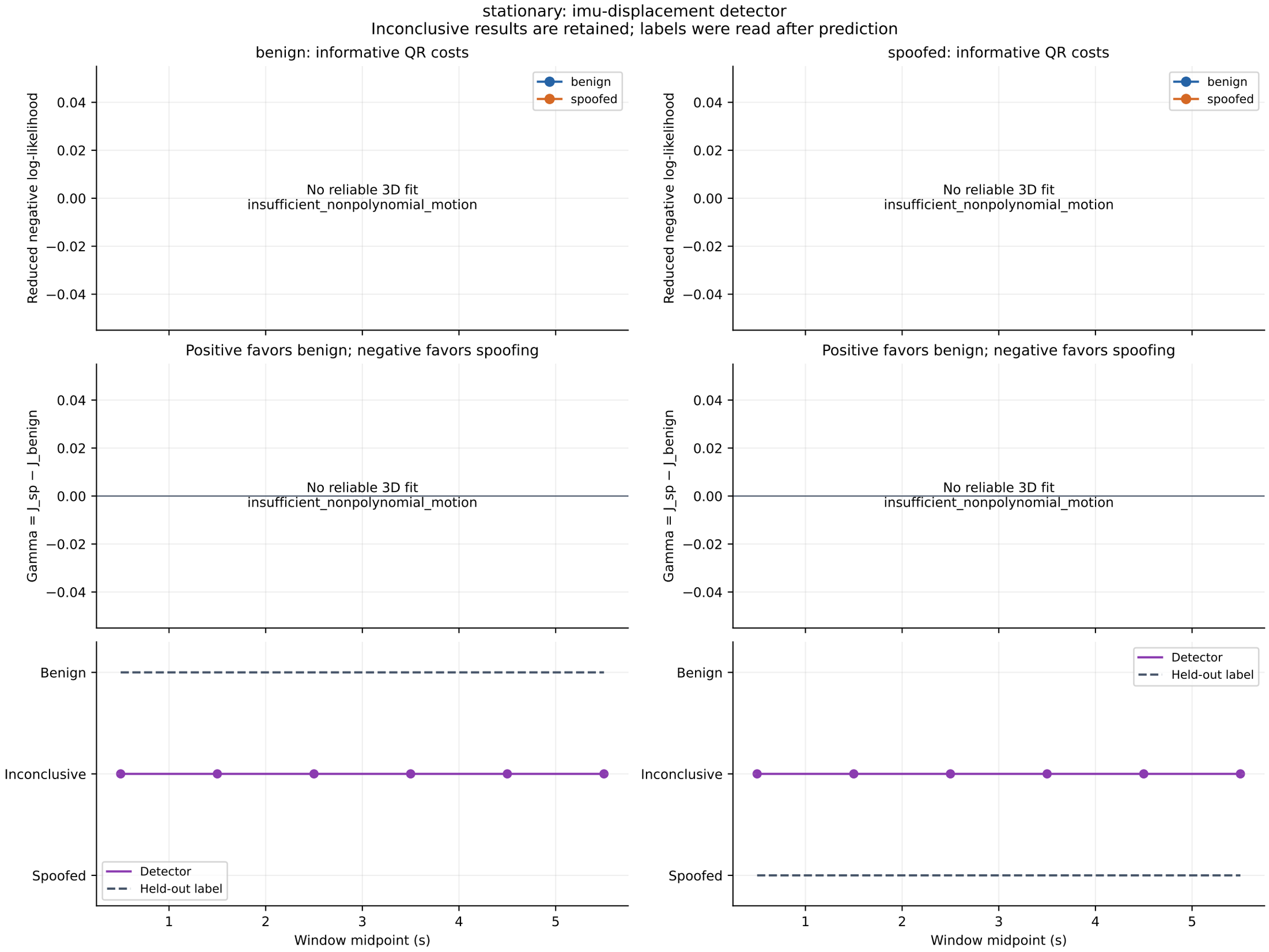
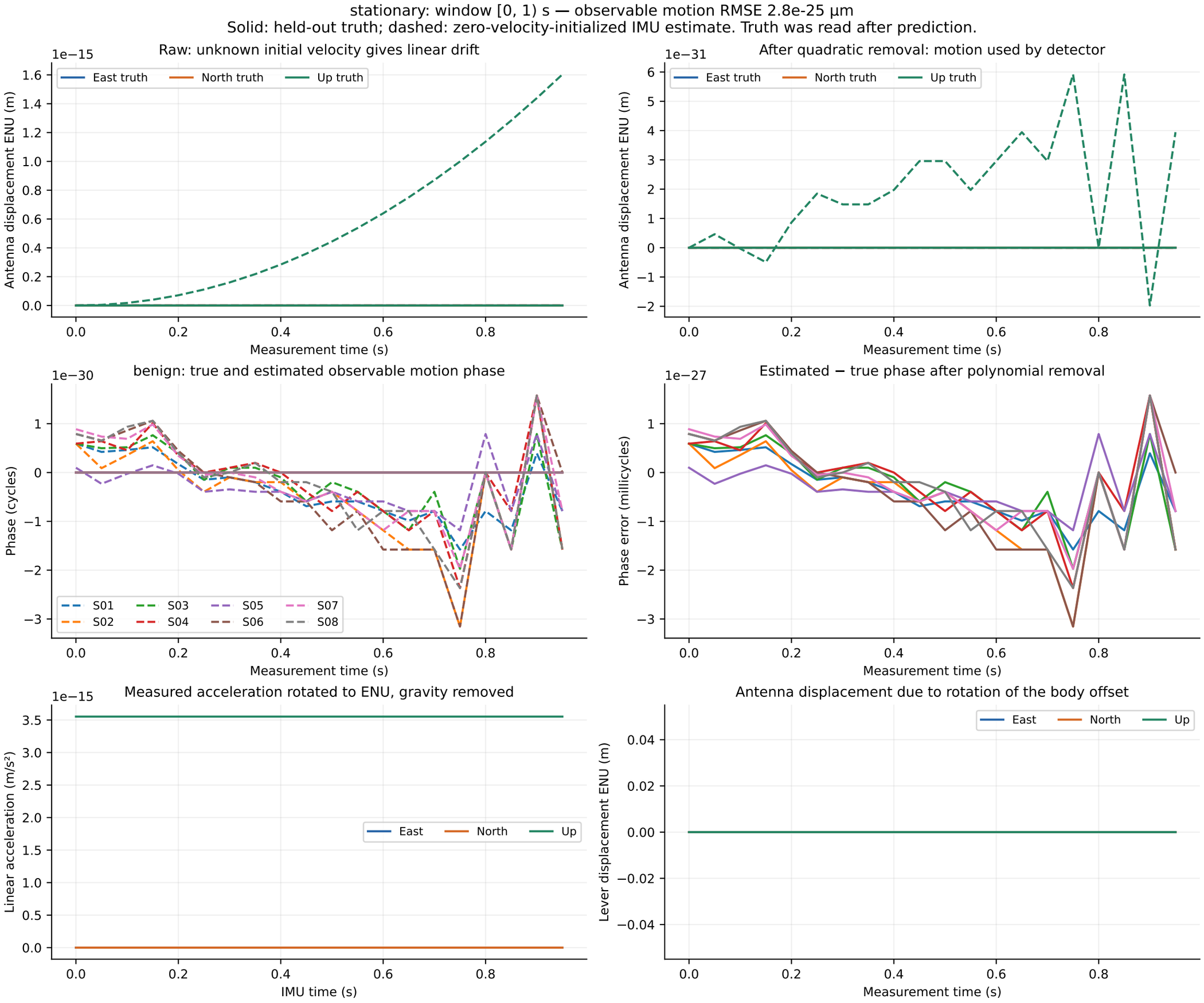
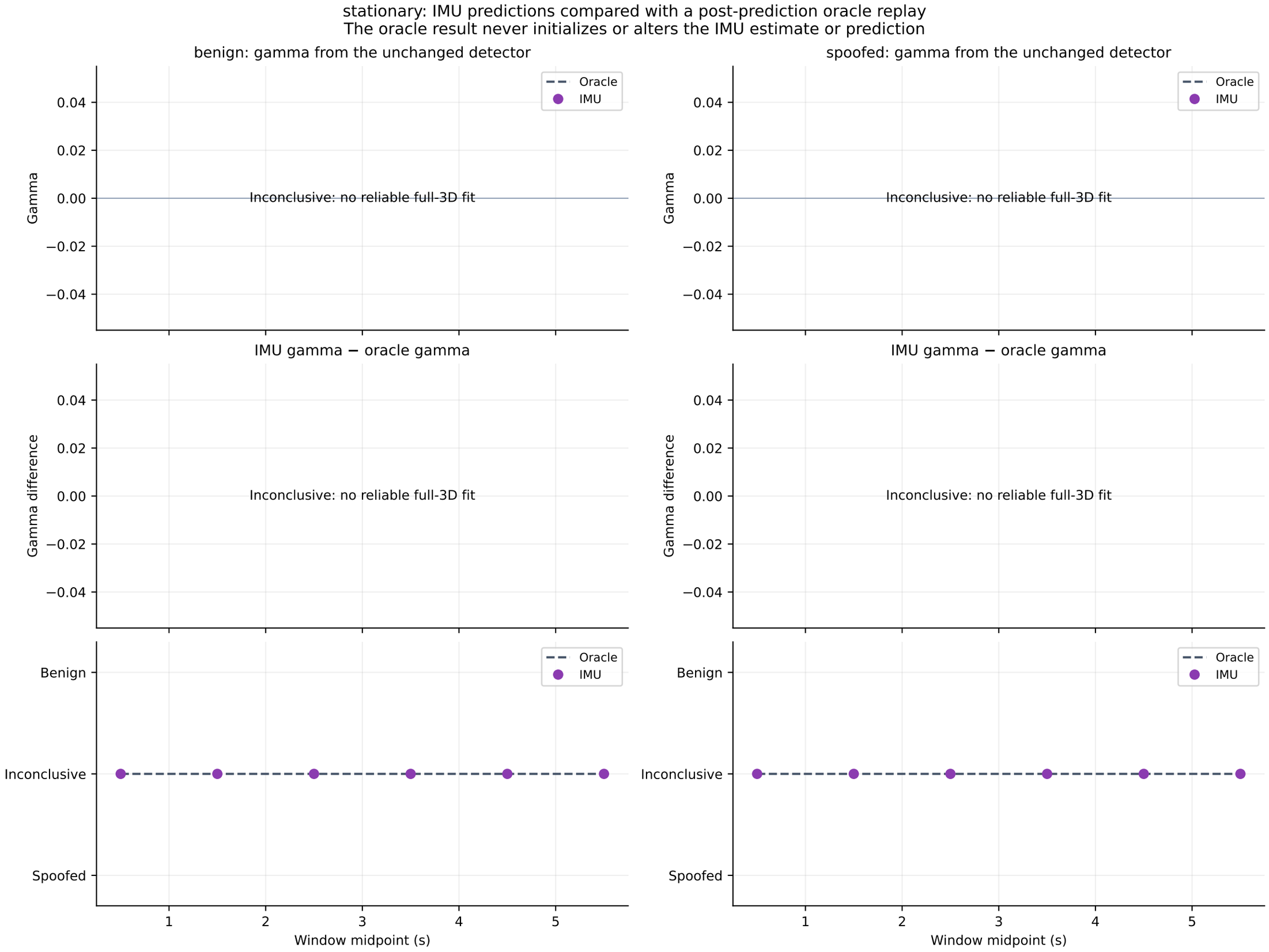
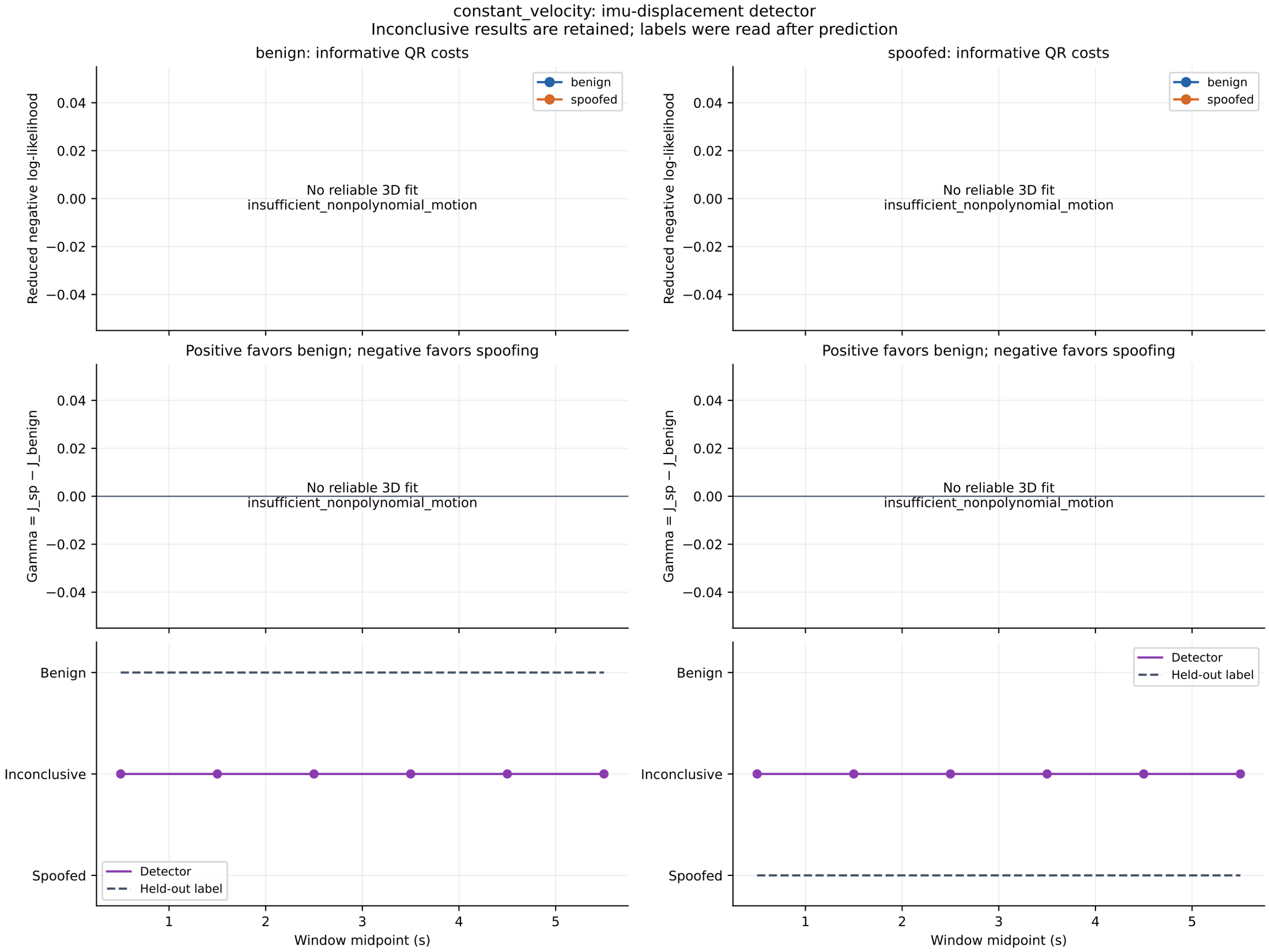
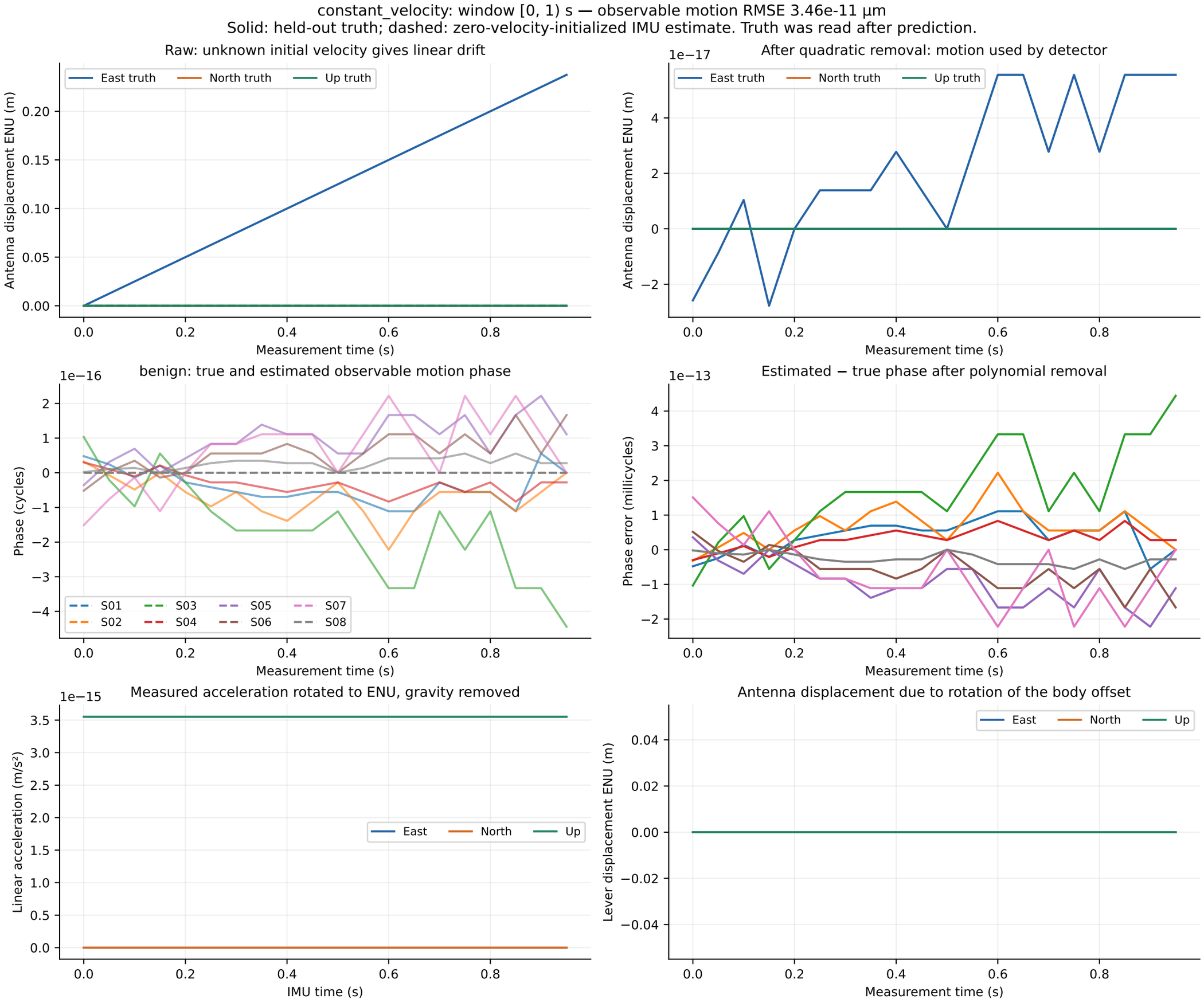
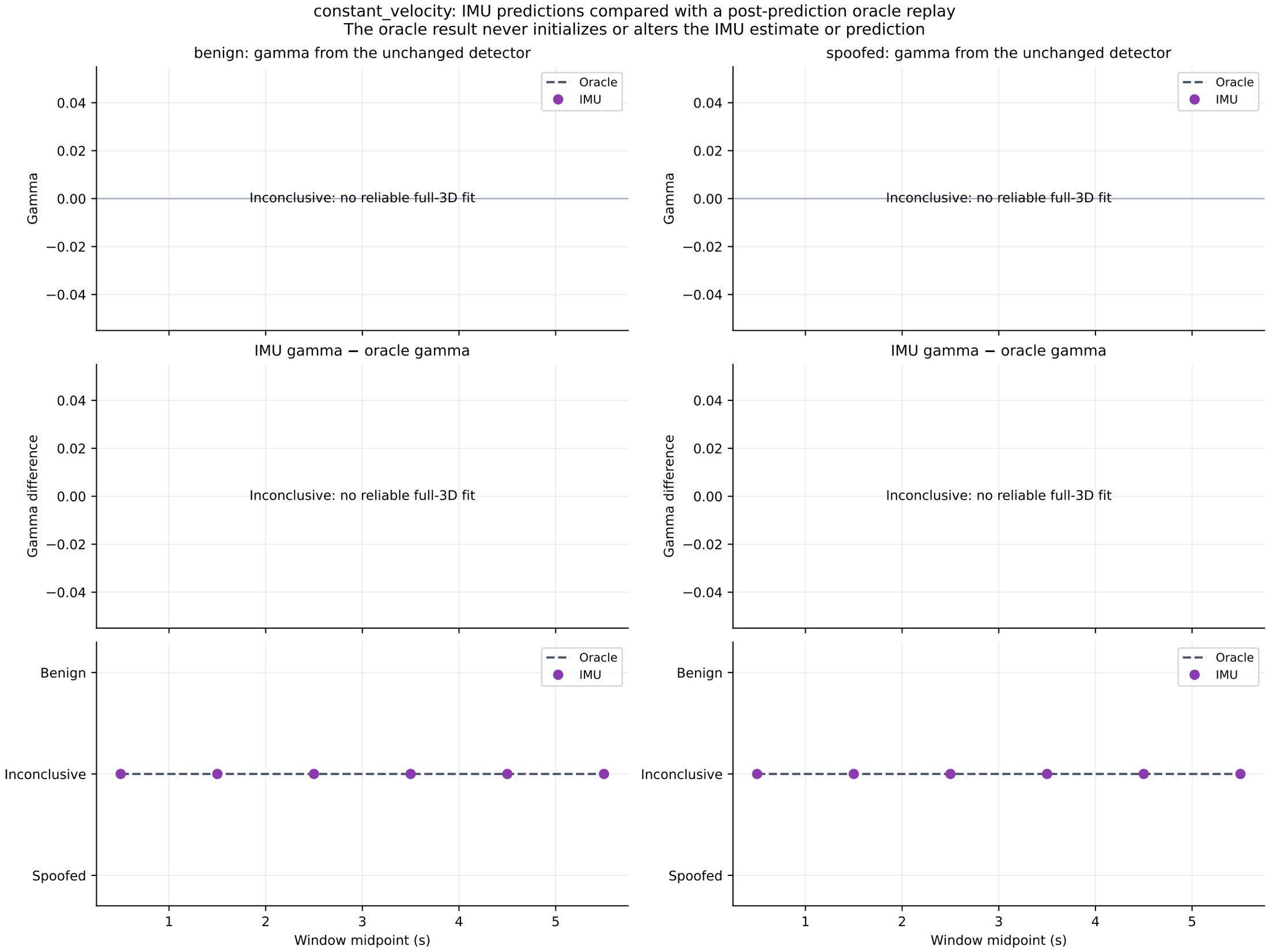

stationary: imu-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
No reliable 3D fit
insufficient_nonpolynomial_motion
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

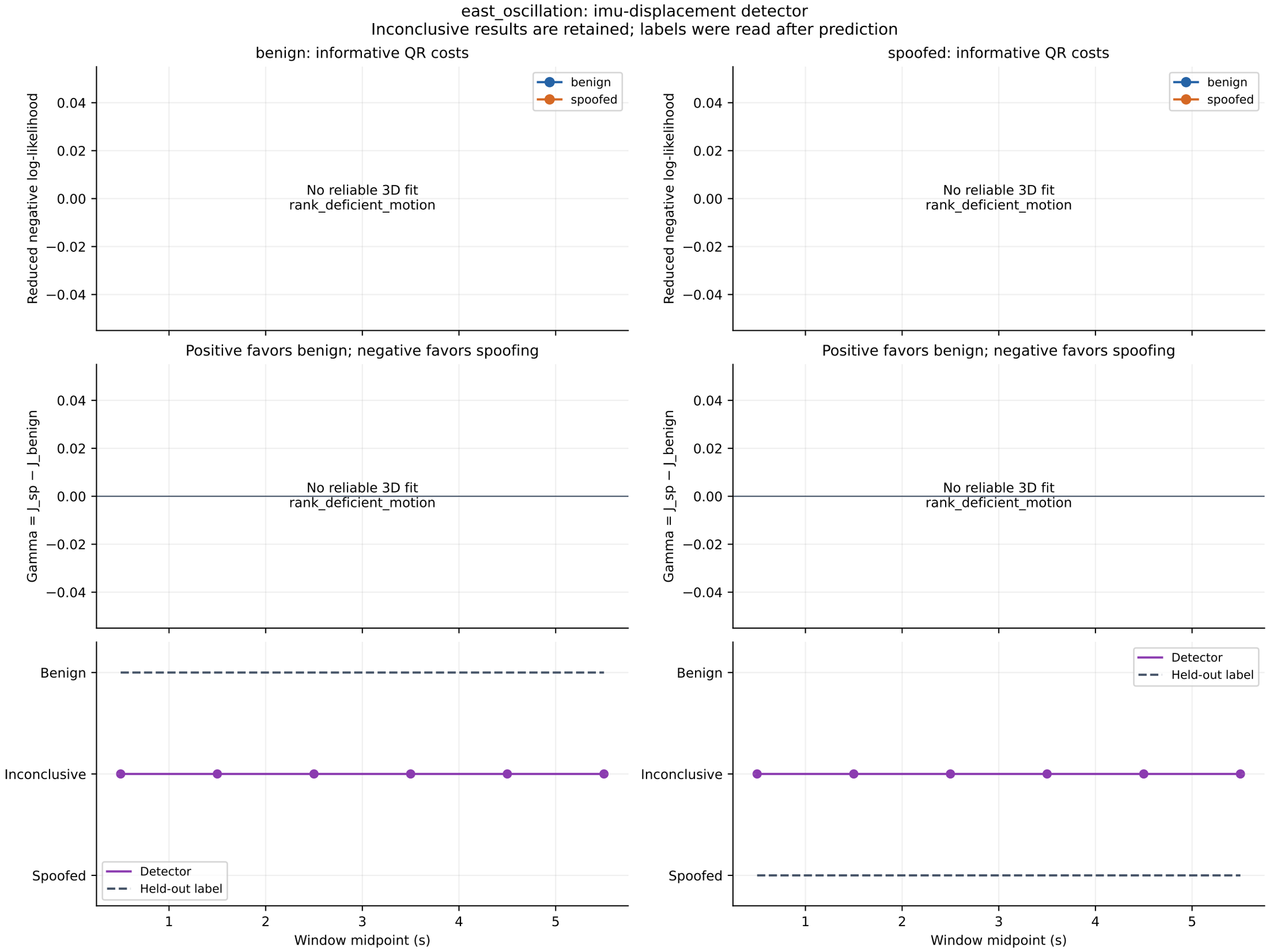

stationary: window [0, 1) s — observable motion RMSE 2.8e-25 µm
Solid: held-out truth; dashed: zero-velocity-initialized IMU estimate. Truth was read after prediction.
Raw: unknown initial velocity gives linear drift
1e−15
Antenna displacement ENU (m)
East truth
North truth
Up truth
Measurement time (s)
After quadratic removal: motion used by detector
1e−31
benign: true and estimated observable motion phase
1e−30
Phase (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Estimated − true phase after polynomial removal
1e−27
Phase error (millicycles)
Measured acceleration rotated to ENU, gravity removed
Linear acceleration (m/s²)
East
North
Up
IMU time (s)
Antenna displacement due to rotation of the body offset
Lever displacement ENU (m)

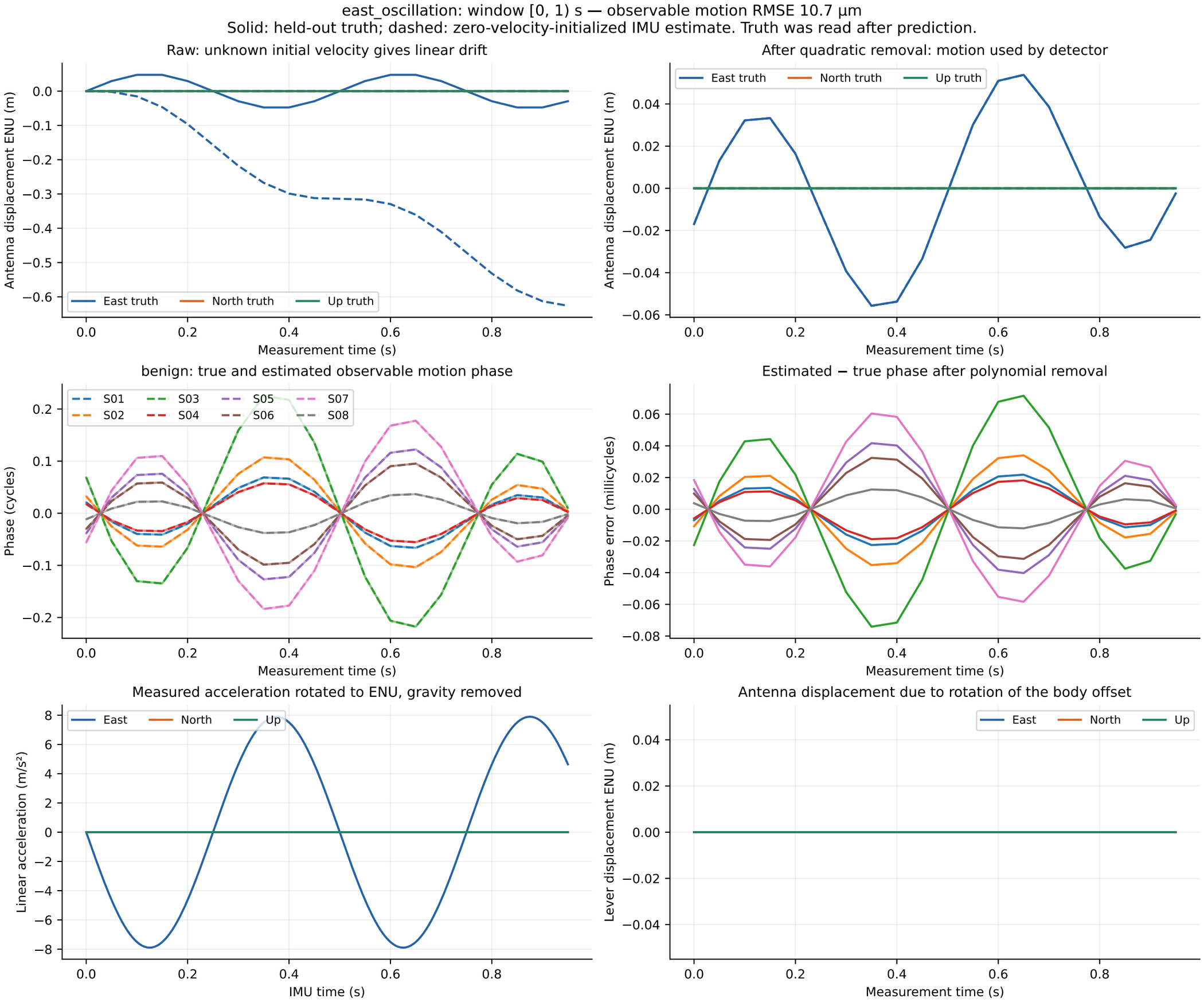

stationary: IMU predictions compared with a post-prediction oracle replay
The oracle result never initializes or alters the IMU estimate or prediction
benign: gamma from the unchanged detector
spoofed: gamma from the unchanged detector
Oracle
IMU
Gamma
0.04
0.02
0.00
−0.02
−0.04
Inconclusive: no reliable full-3D fit
IMU gamma − oracle gamma
Gamma difference
Benign
Inconclusive
Spoofed
1
2
3
4
5
Window midpoint (s)

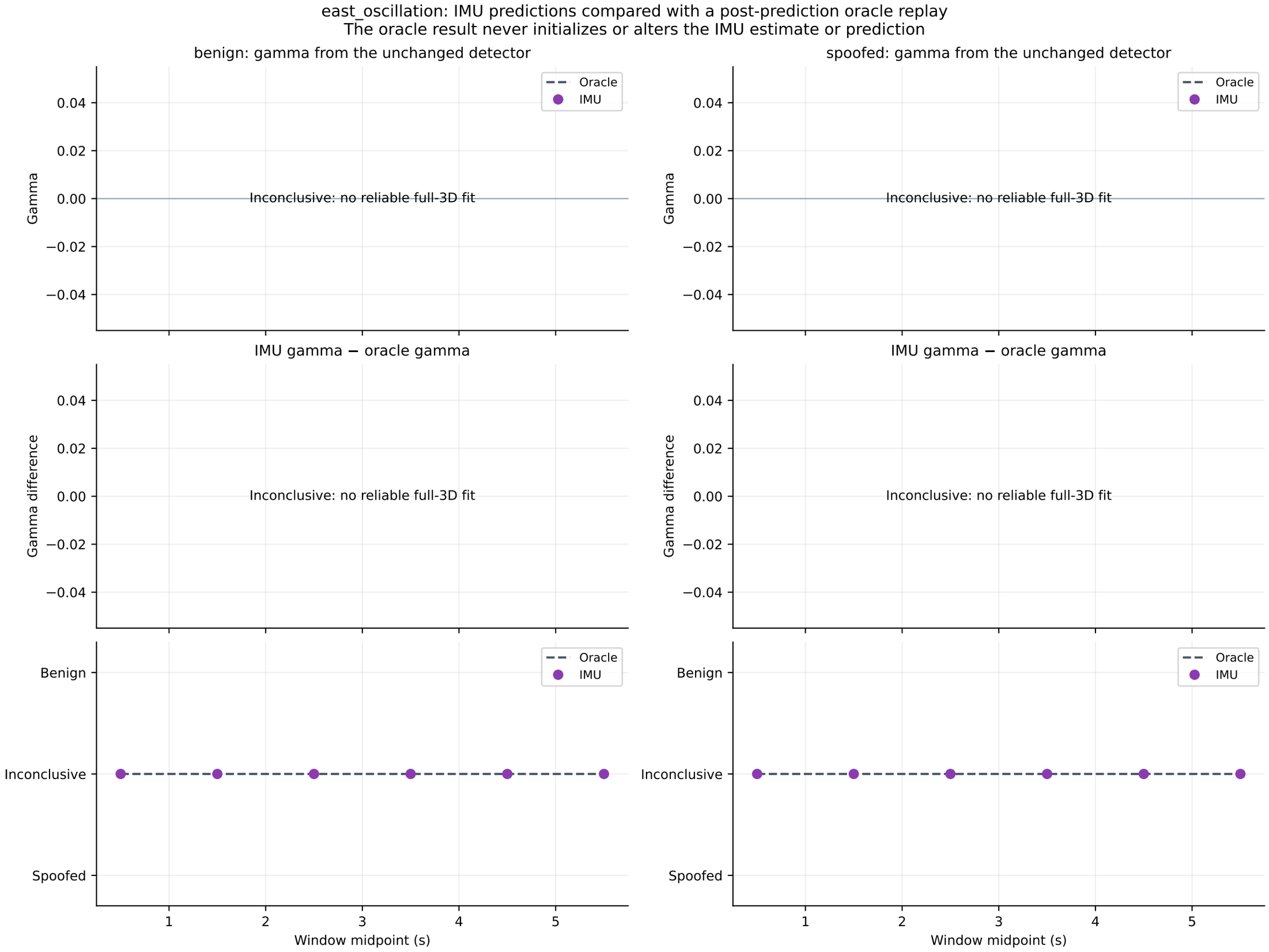

constant_velocity: imu-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
No reliable 3D fit
insufficient_nonpolynomial_motion
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

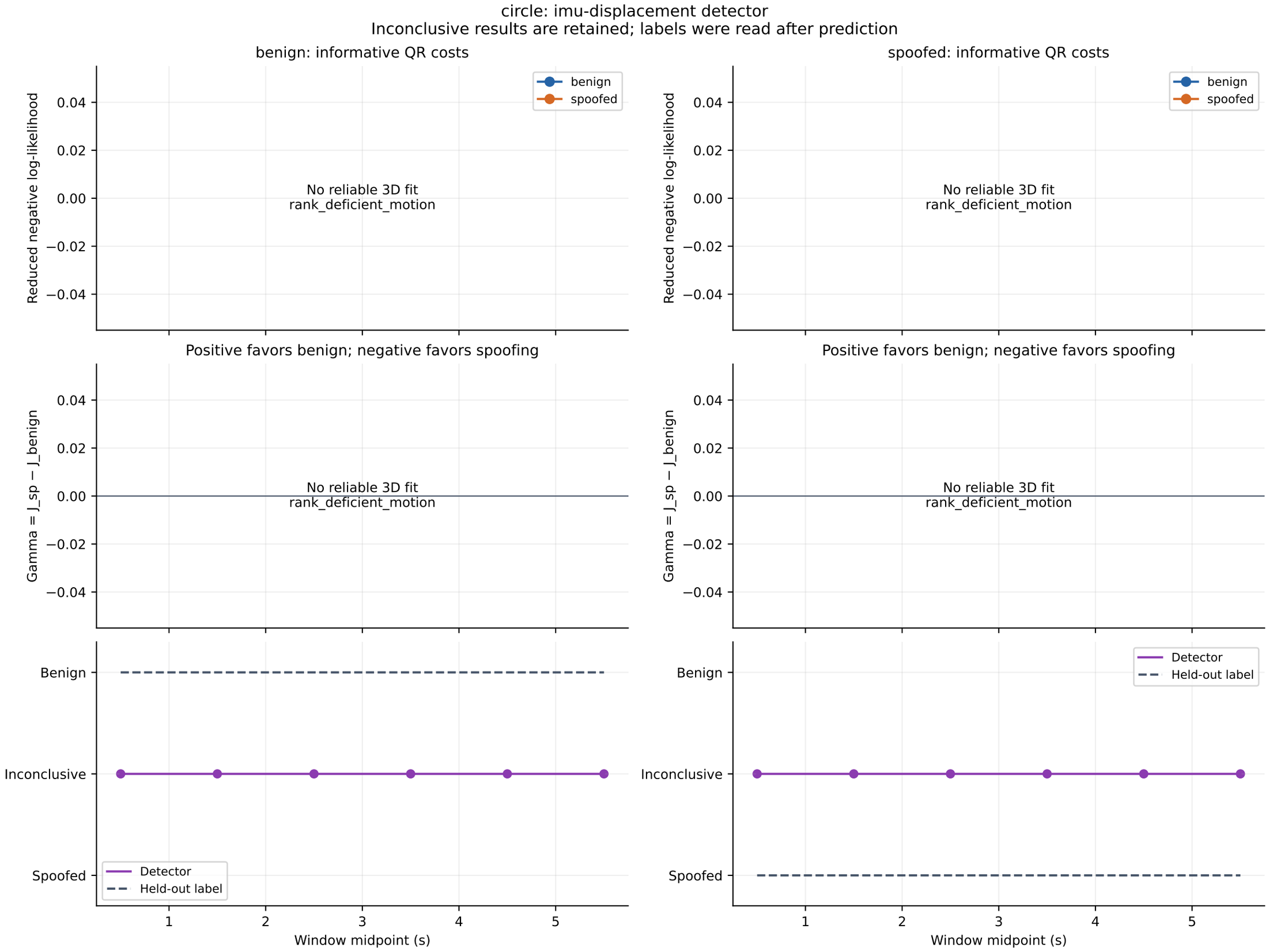

constant_velocity: window [0, 1) s — observable motion RMSE 3.46e-11 µm
Solid: held-out truth; dashed: zero-velocity-initialized IMU estimate. Truth was read after prediction.
Raw: unknown initial velocity gives linear drift
East truth
North truth
Up truth
Antenna displacement ENU (m)
Measurement time (s)
After quadratic removal: motion used by detector
1e−17
benign: true and estimated observable motion phase
1e−16
Phase (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Estimated − true phase after polynomial removal
1e−13
Phase error (millicycles)
Measured acceleration rotated to ENU, gravity removed
1e−15
Linear acceleration (m/s²)
East
North
Up
IMU time (s)
Antenna displacement due to rotation of the body offset
Lever displacement ENU (m)

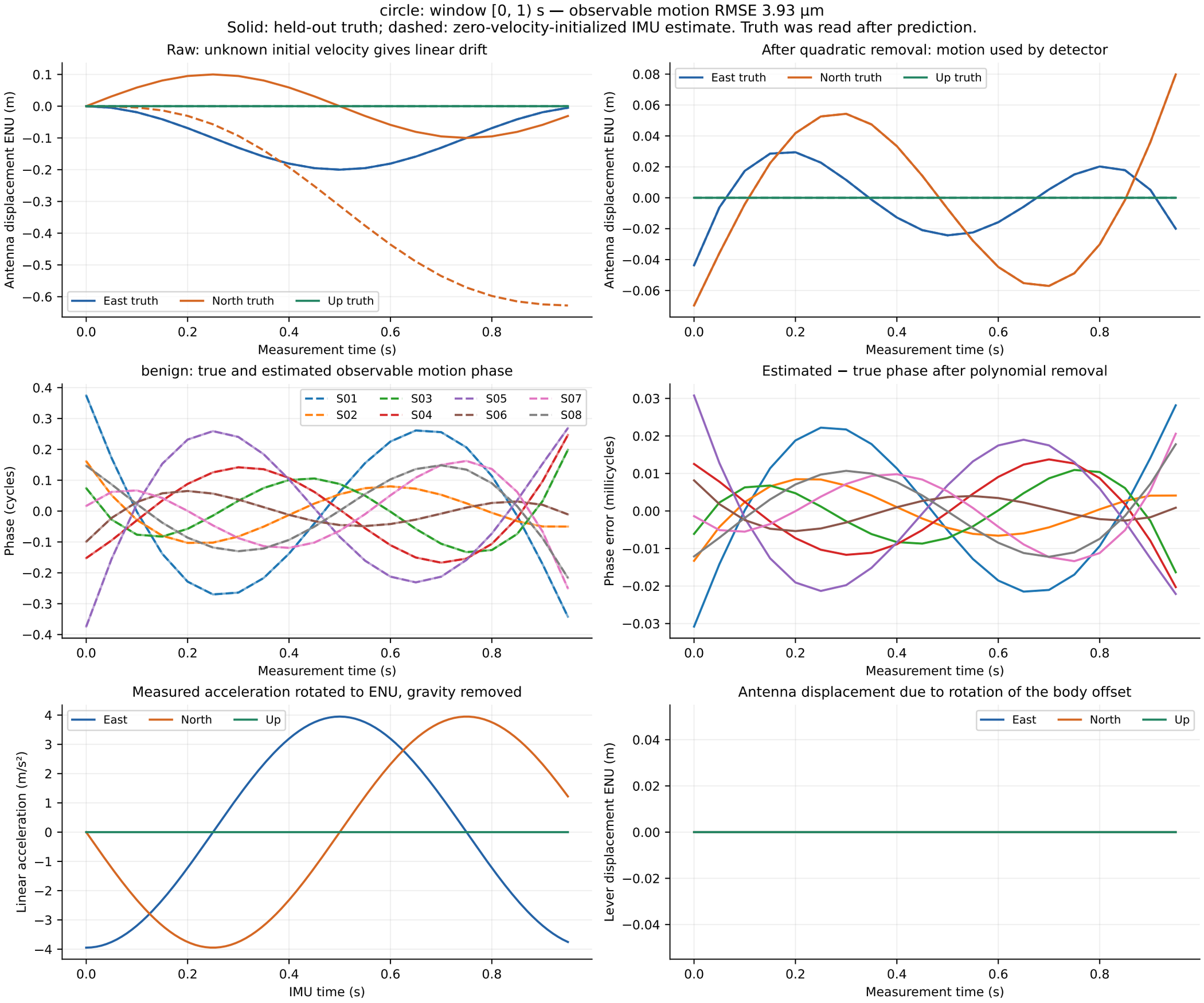

constant_velocity: IMU predictions compared with a post-prediction oracle replay
The oracle result never initializes or alters the IMU estimate or prediction
benign: gamma from the unchanged detector
spoofed: gamma from the unchanged detector
Oracle
IMU
Gamma
0.04
0.02
0.00
−0.02
−0.04
Inconclusive: no reliable full-3D fit
IMU gamma − oracle gamma
Gamma difference
Benign
Inconclusive
Spoofed
1
2
3
4
5
Window midpoint (s)

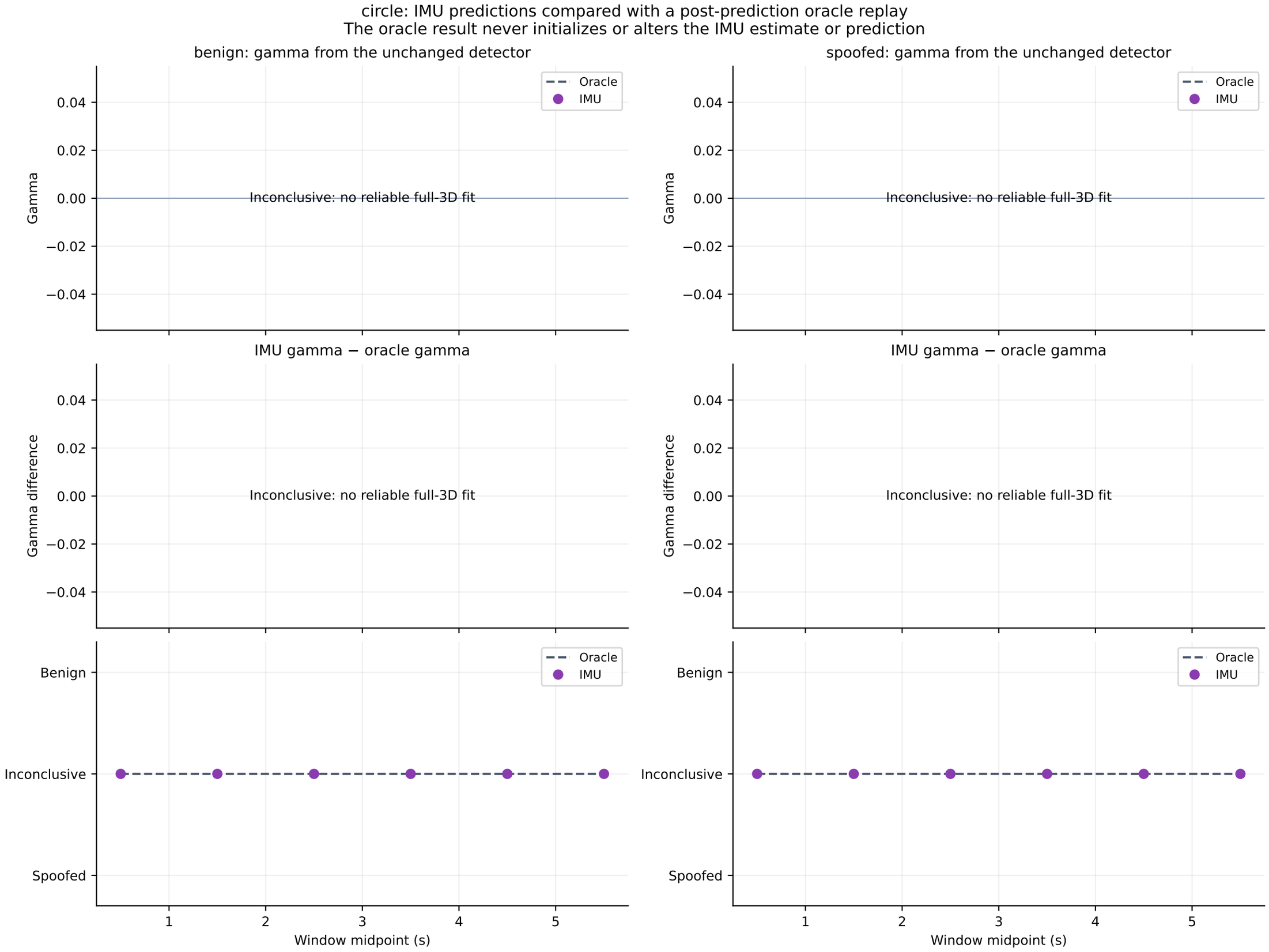

east_oscillation: imu-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
No reliable 3D fit
rank_deficient_motion
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

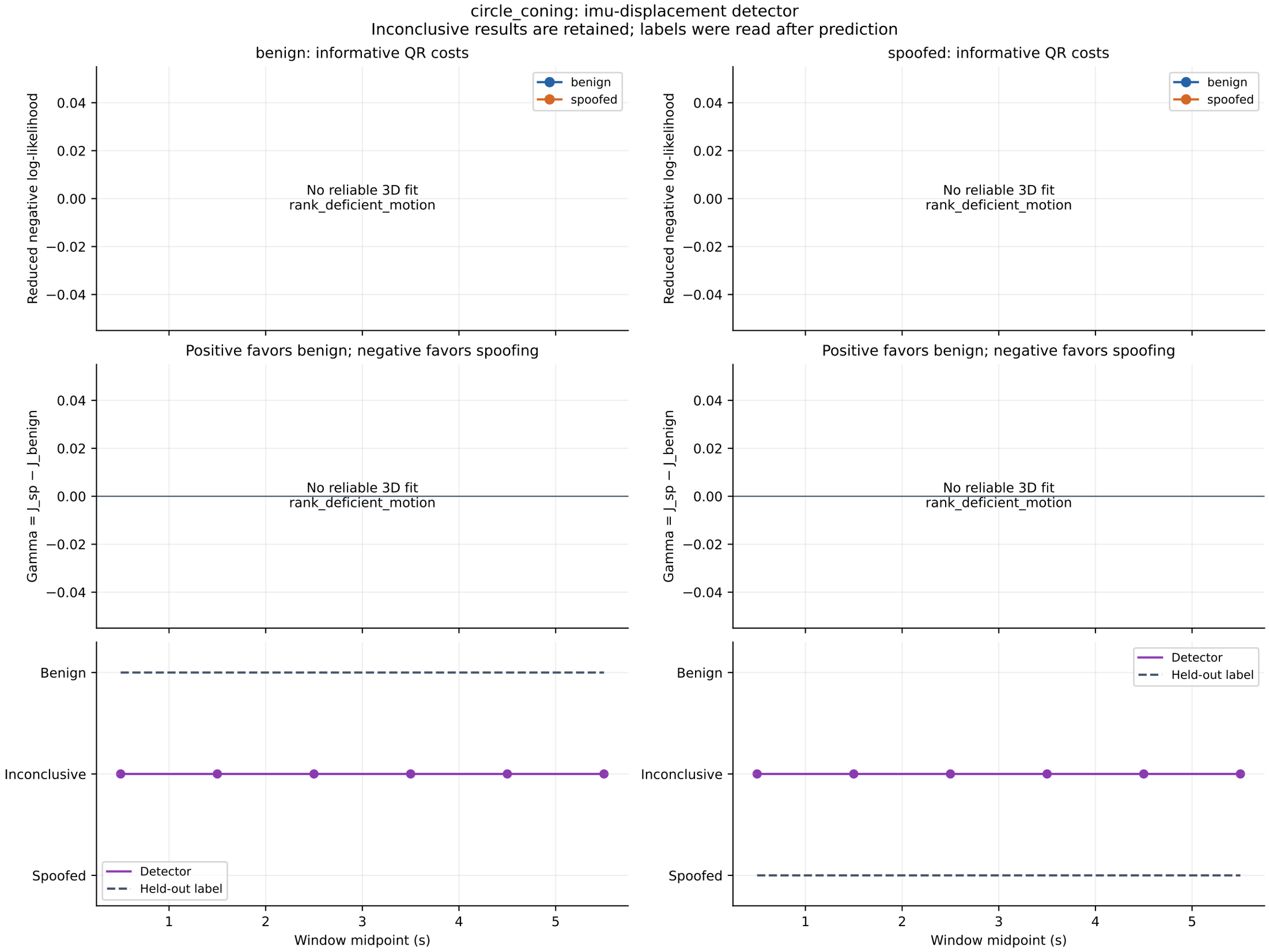

east_oscillation: window [0, 1) s — observable motion RMSE 10.7 µm
Solid: held-out truth; dashed: zero-velocity-initialized IMU estimate. Truth was read after prediction.
Raw: unknown initial velocity gives linear drift
Antenna displacement ENU (m)
East truth
North truth
Up truth
Measurement time (s)
After quadratic removal: motion used by detector
benign: true and estimated observable motion phase
Phase (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Estimated − true phase after polynomial removal
Phase error (millicycles)
Measured acceleration rotated to ENU, gravity removed
Linear acceleration (m/s²)
East
North
Up
IMU time (s)
Antenna displacement due to rotation of the body offset
Lever displacement ENU (m)

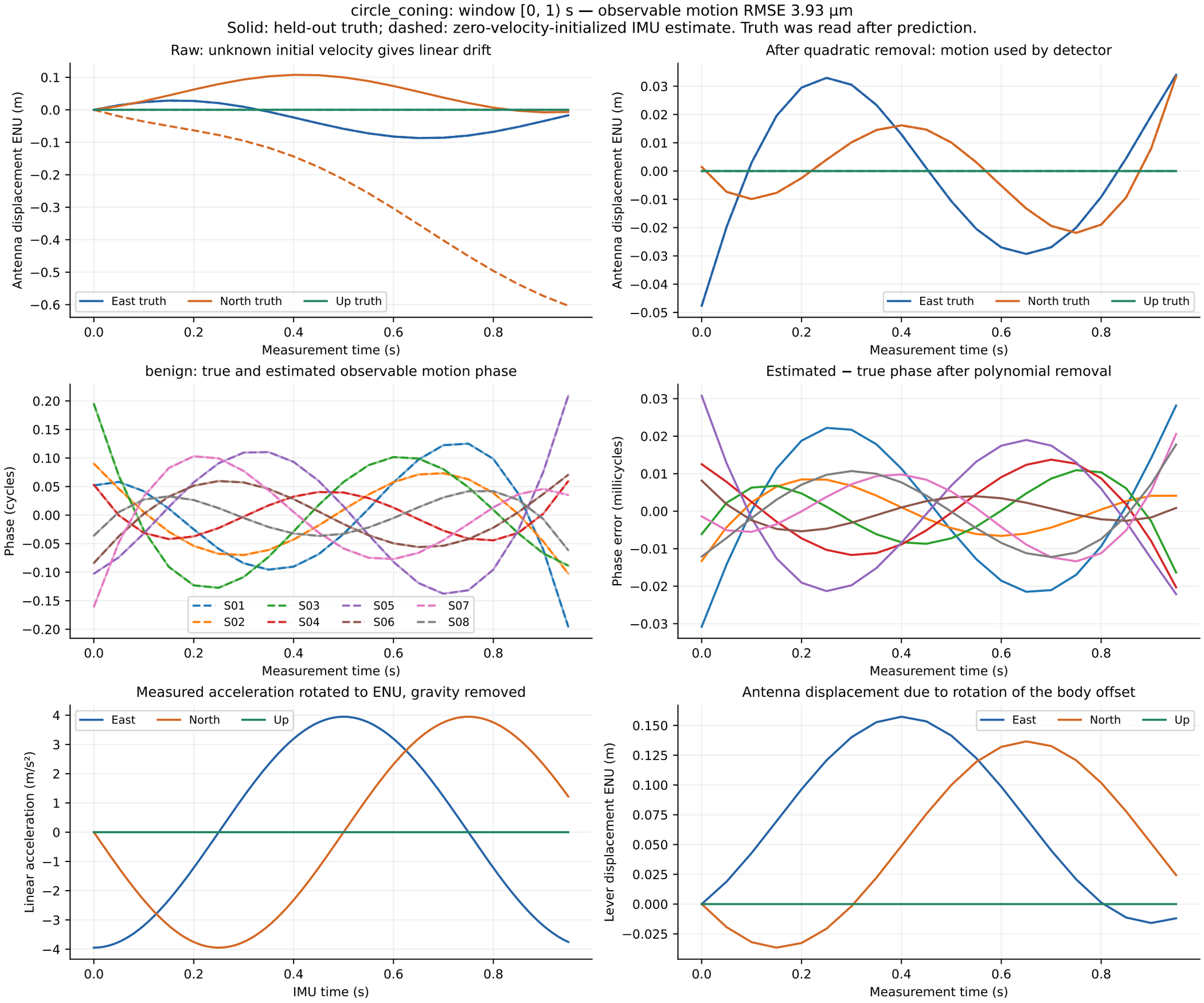

east_oscillation: IMU predictions compared with a post-prediction oracle replay
The oracle result never initializes or alters the IMU estimate or prediction
benign: gamma from the unchanged detector
spoofed: gamma from the unchanged detector
Oracle
IMU
Gamma
0.04
0.02
0.00
−0.02
−0.04
Inconclusive: no reliable full-3D fit
IMU gamma − oracle gamma
Gamma difference
Benign
Inconclusive
Spoofed
1
2
3
4
5
Window midpoint (s)

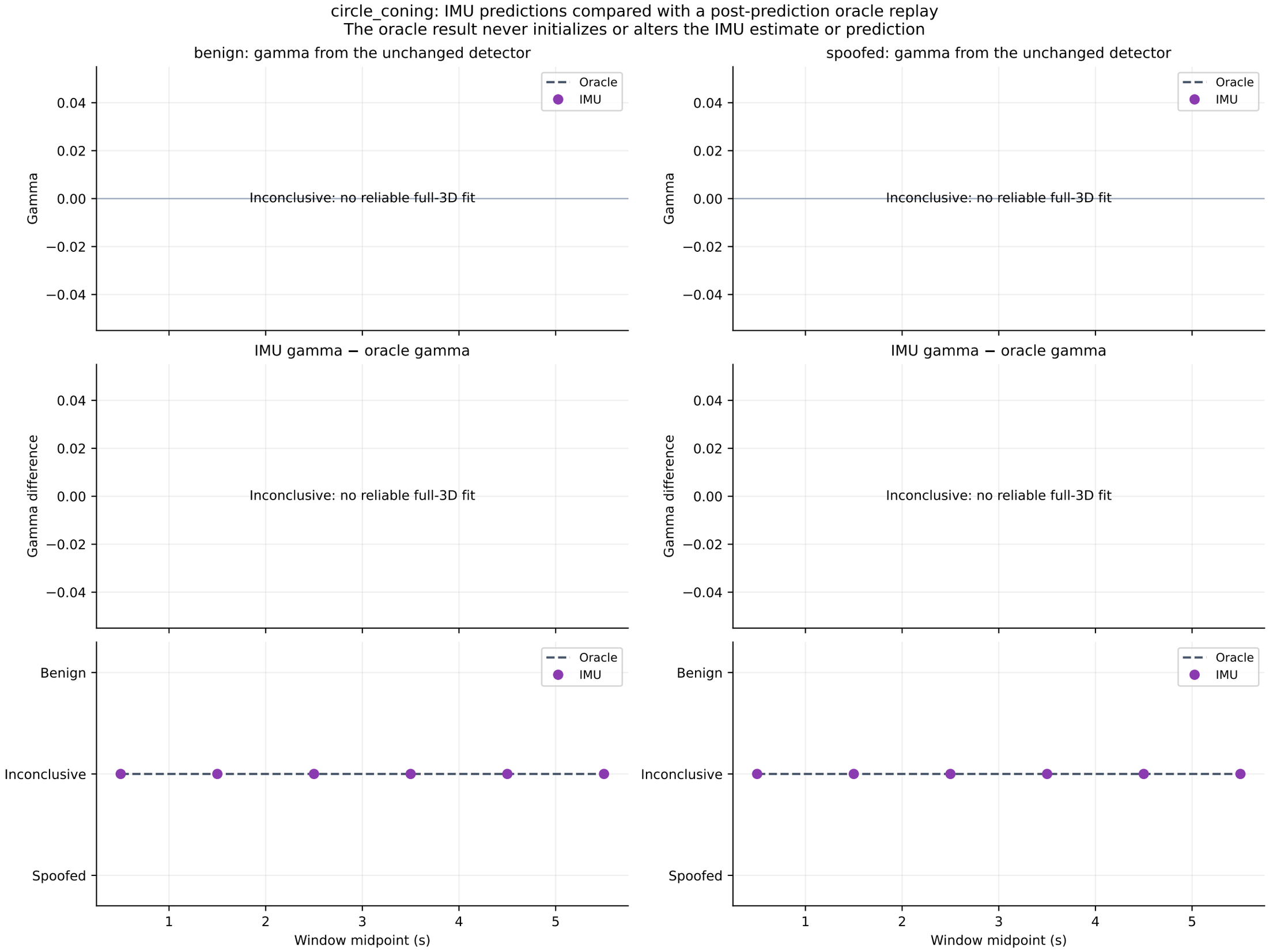

circle: imu-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
No reliable 3D fit
rank_deficient_motion
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

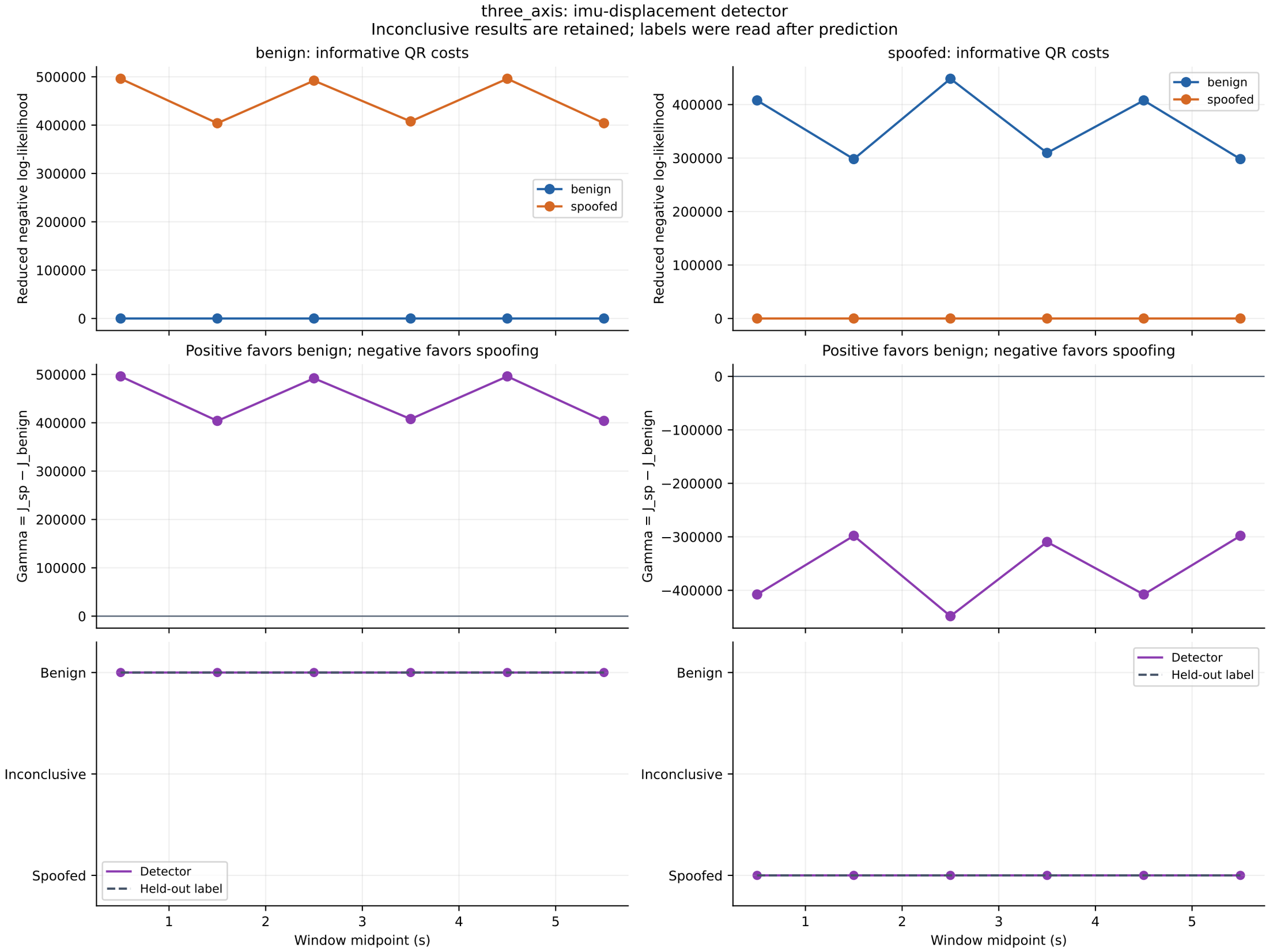

circle: window [0, 1) s — observable motion RMSE 3.93 µm
Solid: held-out truth; dashed: zero-velocity-initialized IMU estimate. Truth was read after prediction.
Raw: unknown initial velocity gives linear drift
Antenna displacement ENU (m)
East truth
North truth
Up truth
Measurement time (s)
After quadratic removal: motion used by detector
Antenna displacement ENU (m)
East truth
North truth
Up truth
Measurement time (s)
benign: true and estimated observable motion phase
Phase (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Measurement time (s)
Estimated − true phase after polynomial removal
Phase error (millicycles)
Measurement time (s)
Measured acceleration rotated to ENU, gravity removed
Linear acceleration (m/s²)
East
North
Up
IMU time (s)
Antenna displacement due to rotation of the body offset
Lever displacement ENU (m)
East
North
Up
Measurement time (s)

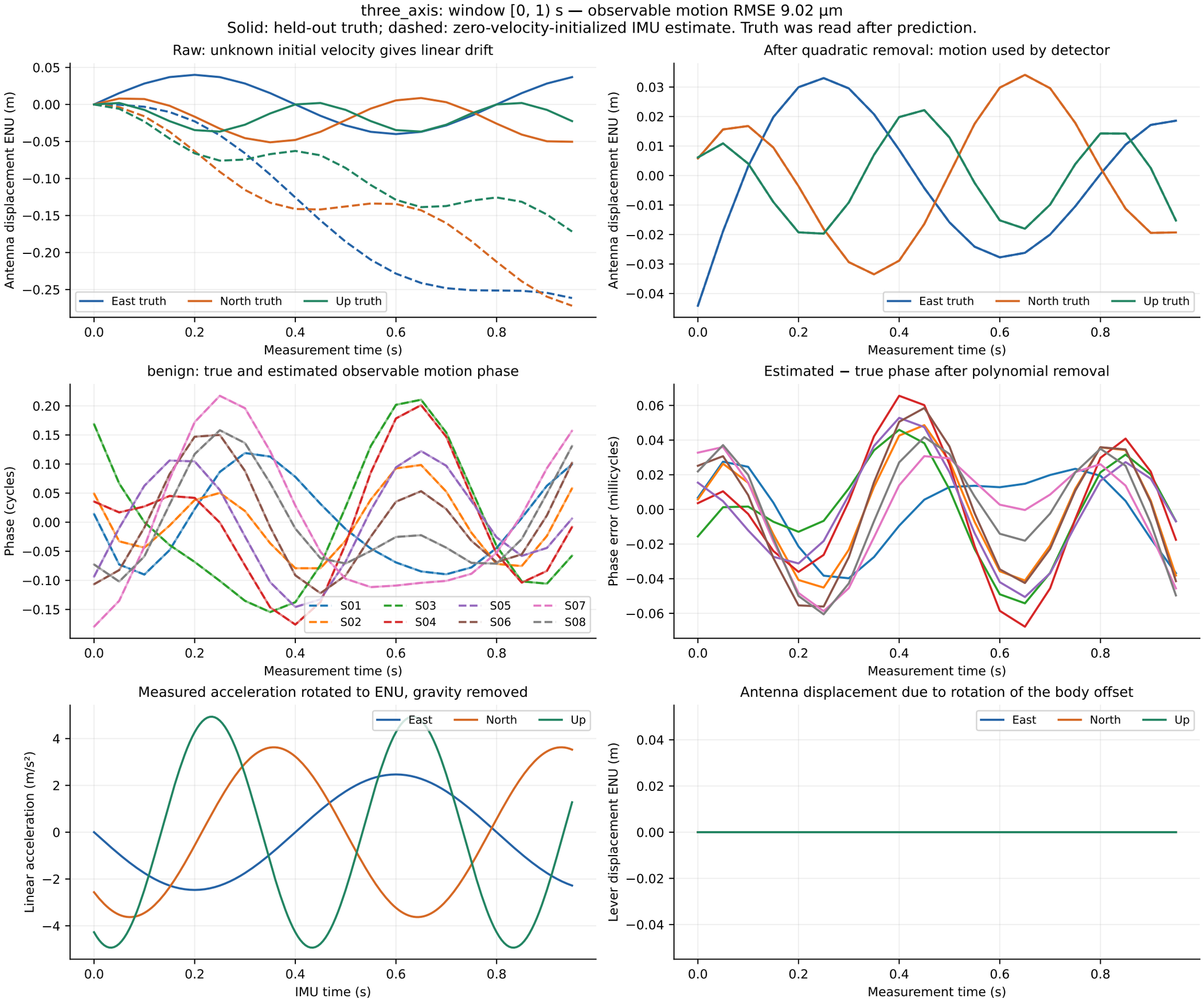

circle: IMU predictions compared with a post-prediction oracle replay
The oracle result never initializes or alters the IMU estimate or prediction
benign: gamma from the unchanged detector
spoofed: gamma from the unchanged detector
Oracle
IMU
Gamma
0.04
0.02
0.00
−0.02
−0.04
Inconclusive: no reliable full-3D fit
IMU gamma − oracle gamma
Gamma difference
Benign
Inconclusive
Spoofed
1
2
3
4
5
Window midpoint (s)

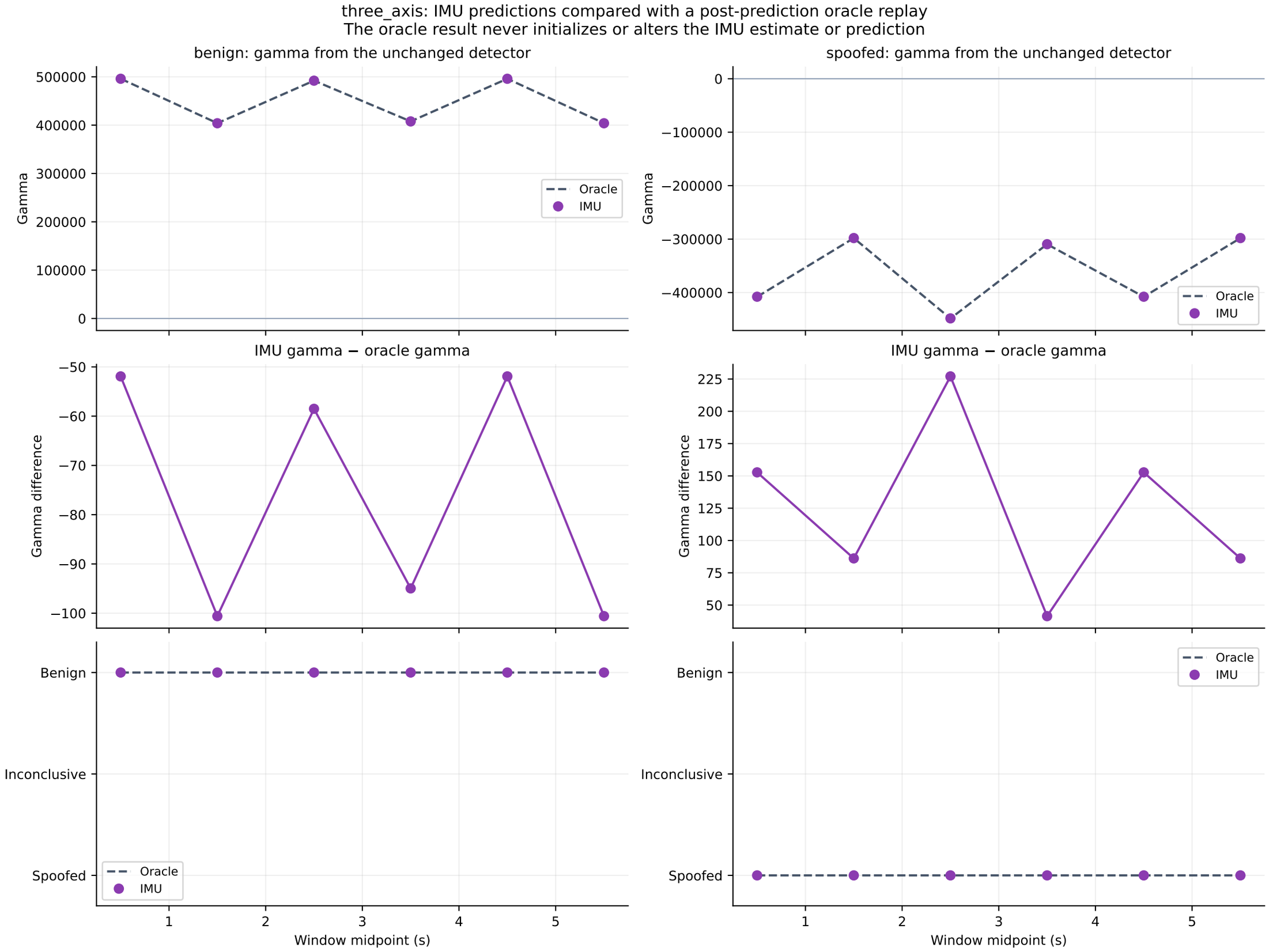

circle_coning: imu-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
No reliable 3D fit
rank_deficient_motion
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

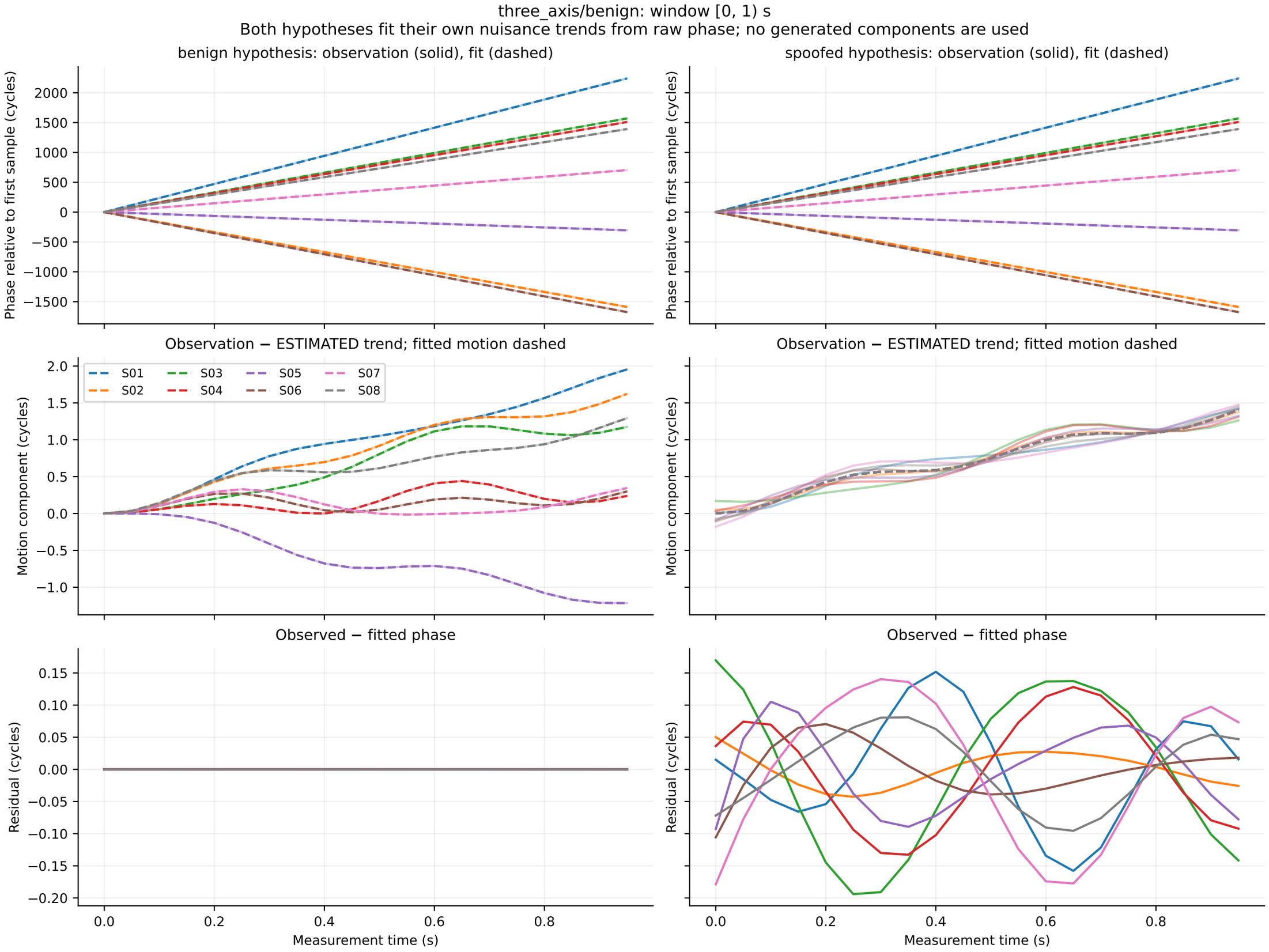

circle_coning: window [0, 1) s — observable motion RMSE 3.93 µm
Solid: held-out truth; dashed: zero-velocity-initialized IMU estimate. Truth was read after prediction.
Raw: unknown initial velocity gives linear drift
Antenna displacement ENU (m)
East truth
North truth
Up truth
Measurement time (s)
After quadratic removal: motion used by detector
Antenna displacement ENU (m)
East truth
North truth
Up truth
Measurement time (s)
benign: true and estimated observable motion phase
Phase (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Measurement time (s)
Estimated − true phase after polynomial removal
Phase error (millicycles)
Measurement time (s)
Measured acceleration rotated to ENU, gravity removed
Linear acceleration (m/s²)
East
North
Up
IMU time (s)
Antenna displacement due to rotation of the body offset
Lever displacement ENU (m)
East
North
Up
Measurement time (s)

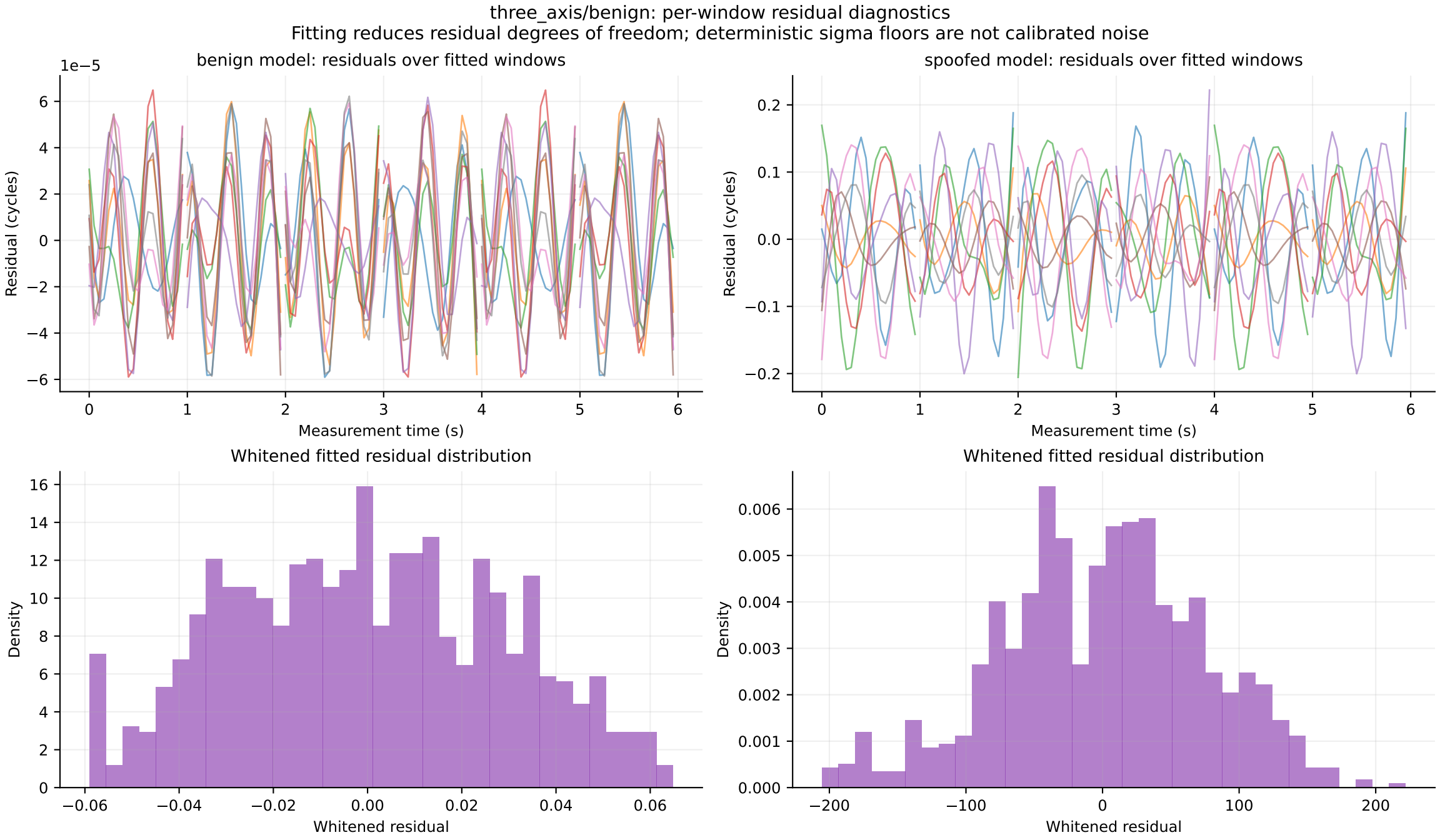

circle_coning: IMU predictions compared with a post-prediction oracle replay
The oracle result never initializes or alters the IMU estimate or prediction
benign: gamma from the unchanged detector
spoofed: gamma from the unchanged detector
Oracle
IMU
Gamma
0.04
0.02
0.00
−0.02
−0.04
Inconclusive: no reliable full-3D fit
IMU gamma − oracle gamma
Gamma difference
Benign
Inconclusive
Spoofed
1
2
3
4
5
Window midpoint (s)

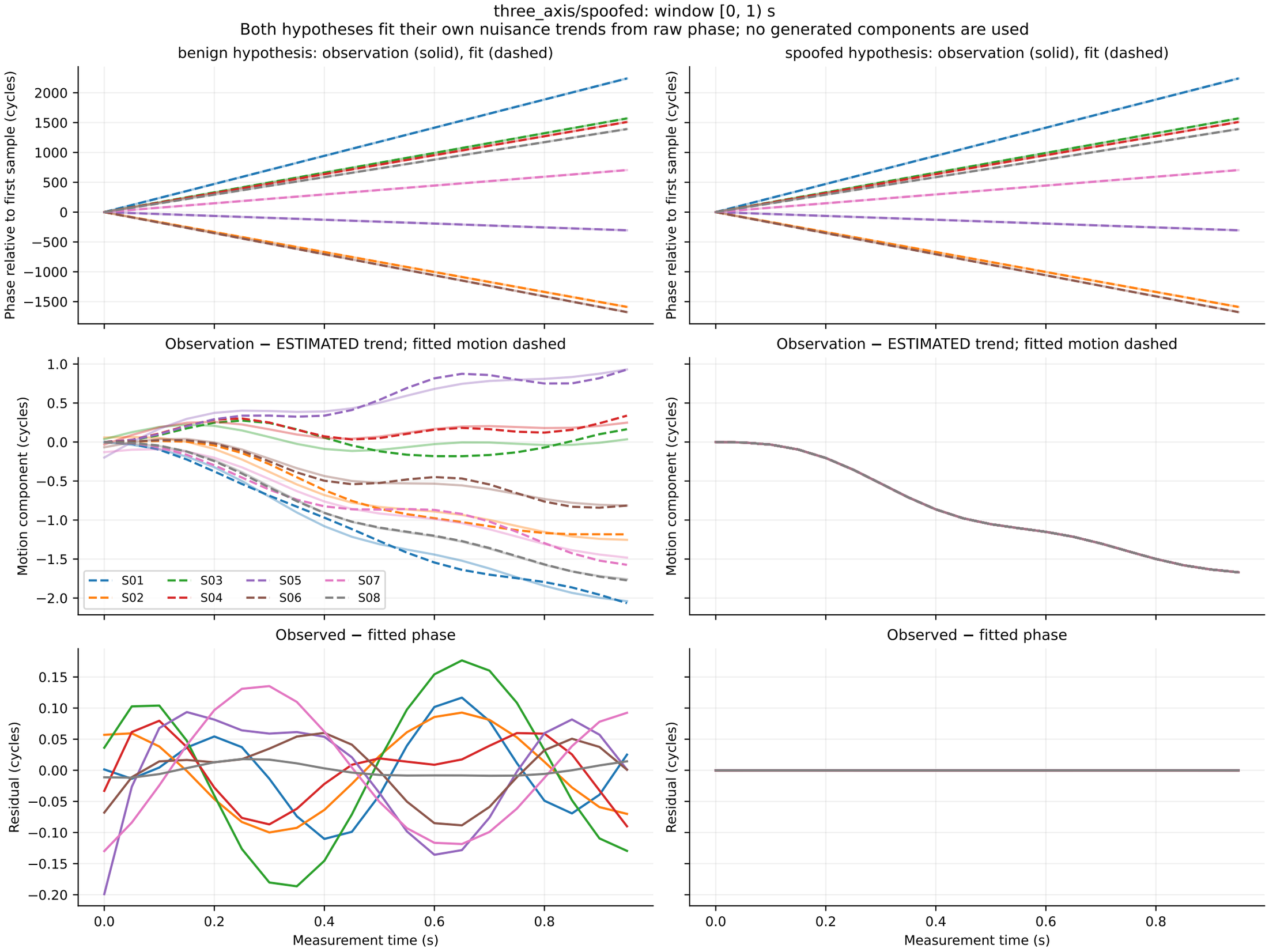

three_axis: imu-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

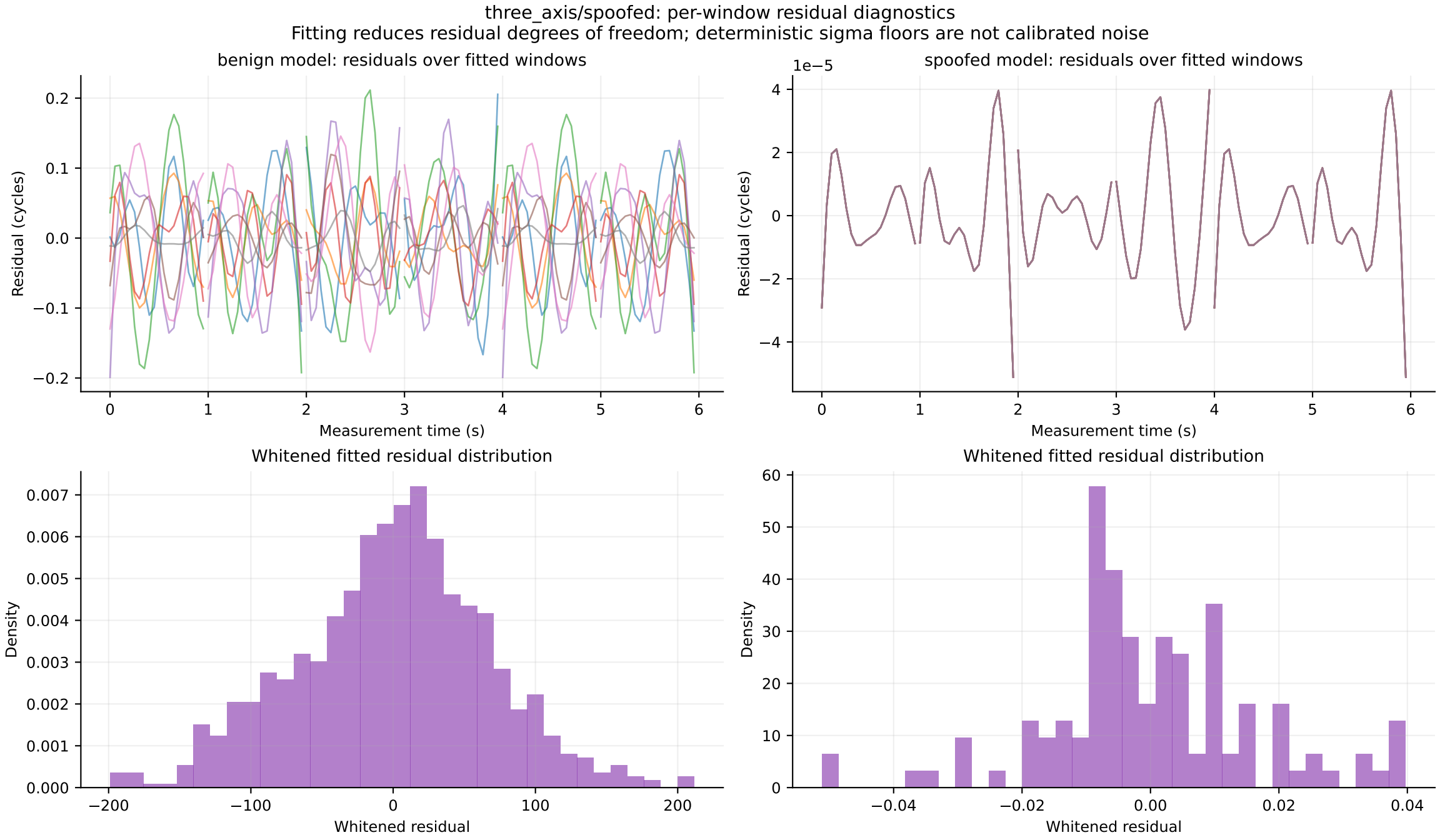

three_axis: window [0, 1) s — observable motion RMSE 9.02 µm
Solid: held-out truth; dashed: zero-velocity-initialized IMU estimate. Truth was read after prediction.
Raw: unknown initial velocity gives linear drift
Antenna displacement ENU (m)
East truth
North truth
Up truth
Measurement time (s)
After quadratic removal: motion used by detector
Antenna displacement ENU (m)
East truth
North truth
Up truth
Measurement time (s)
benign: true and estimated observable motion phase
Phase (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Measurement time (s)
Estimated − true phase after polynomial removal
Phase error (millicycles)
Measurement time (s)
Measured acceleration rotated to ENU, gravity removed
Linear acceleration (m/s²)
East
North
Up
IMU time (s)
Antenna displacement due to rotation of the body offset
Lever displacement ENU (m)
East
North
Up
Measurement time (s)

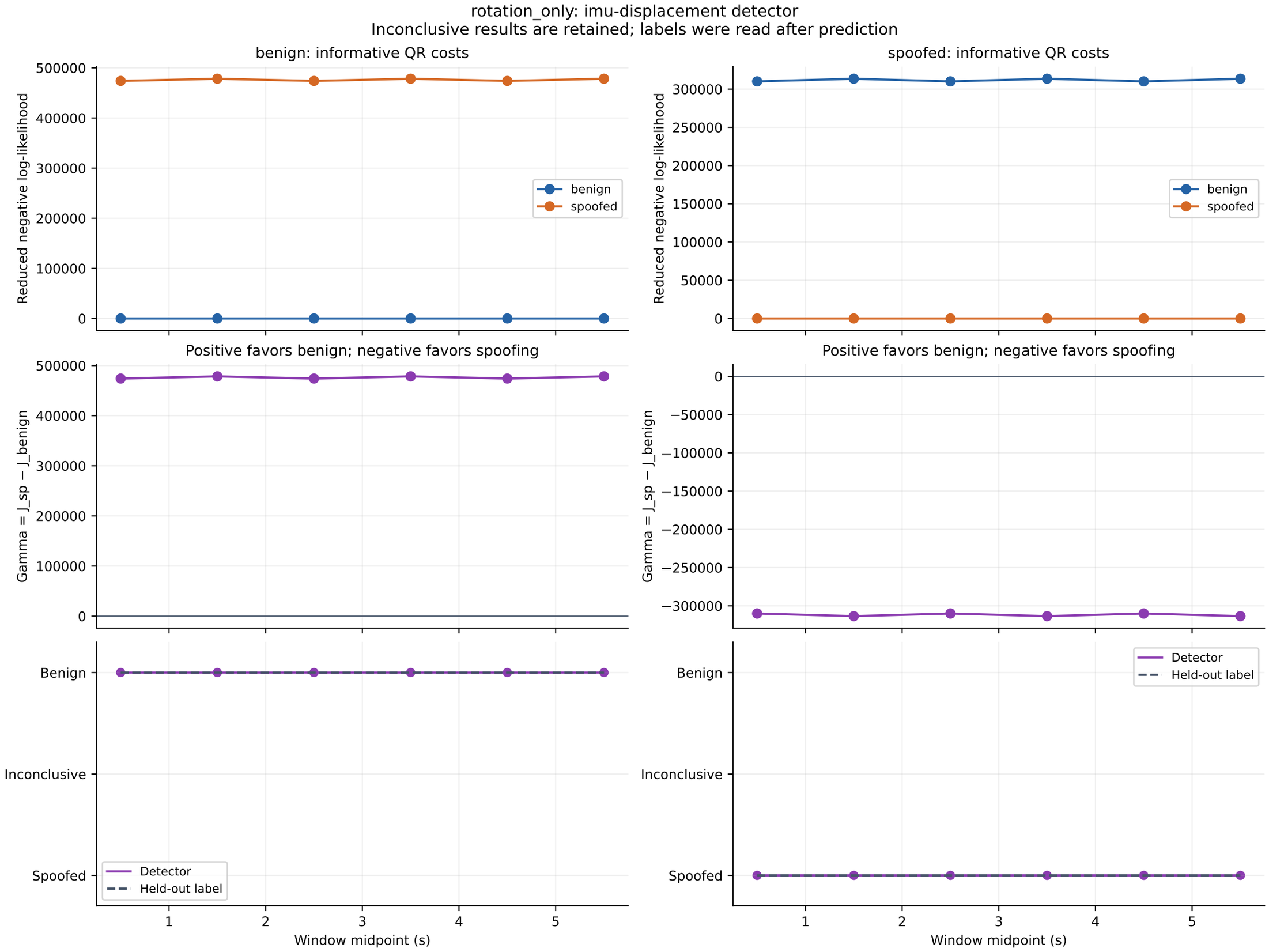

three_axis: IMU predictions compared with a post-prediction oracle replay
The oracle result never initializes or alters the IMU estimate or prediction
benign: gamma from the unchanged detector
spoofed: gamma from the unchanged detector
Gamma
500000
400000
300000
200000
100000
0
−100000
−200000
−300000
−400000
Oracle
IMU
IMU gamma − oracle gamma
Gamma difference
−50
−60
−70
−80
−90
−100
225
200
175
150
125
100
75
50
Benign
Inconclusive
Spoofed
1
2
3
4
5
Window midpoint (s)

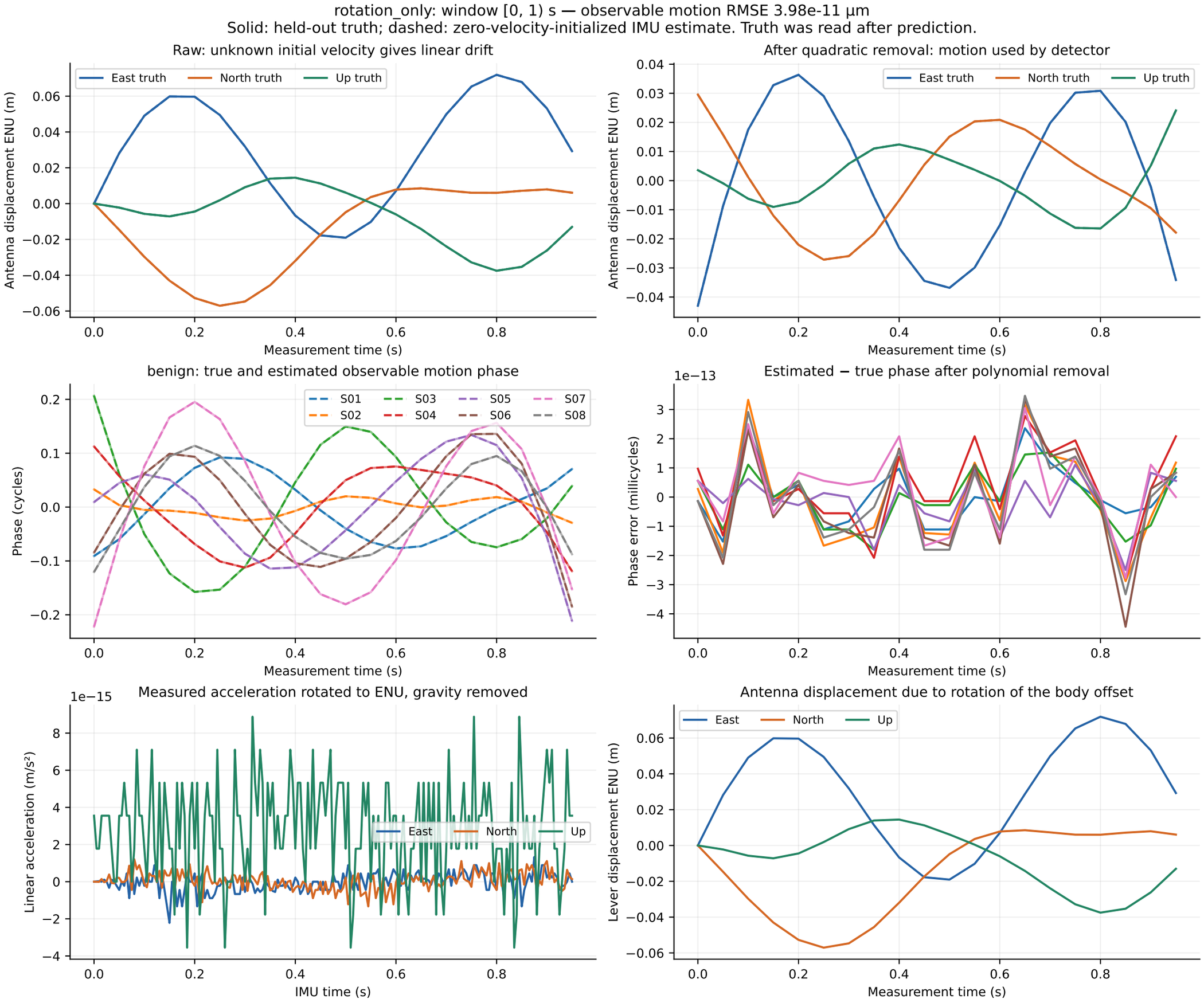

three_axis/benign: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
benign hypothesis: observation (solid), fit (dashed)
spoofed hypothesis: observation (solid), fit (dashed)
Phase relative to first sample (cycles)
Observation − ESTIMATED trend; fitted motion dashed
Motion component (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)

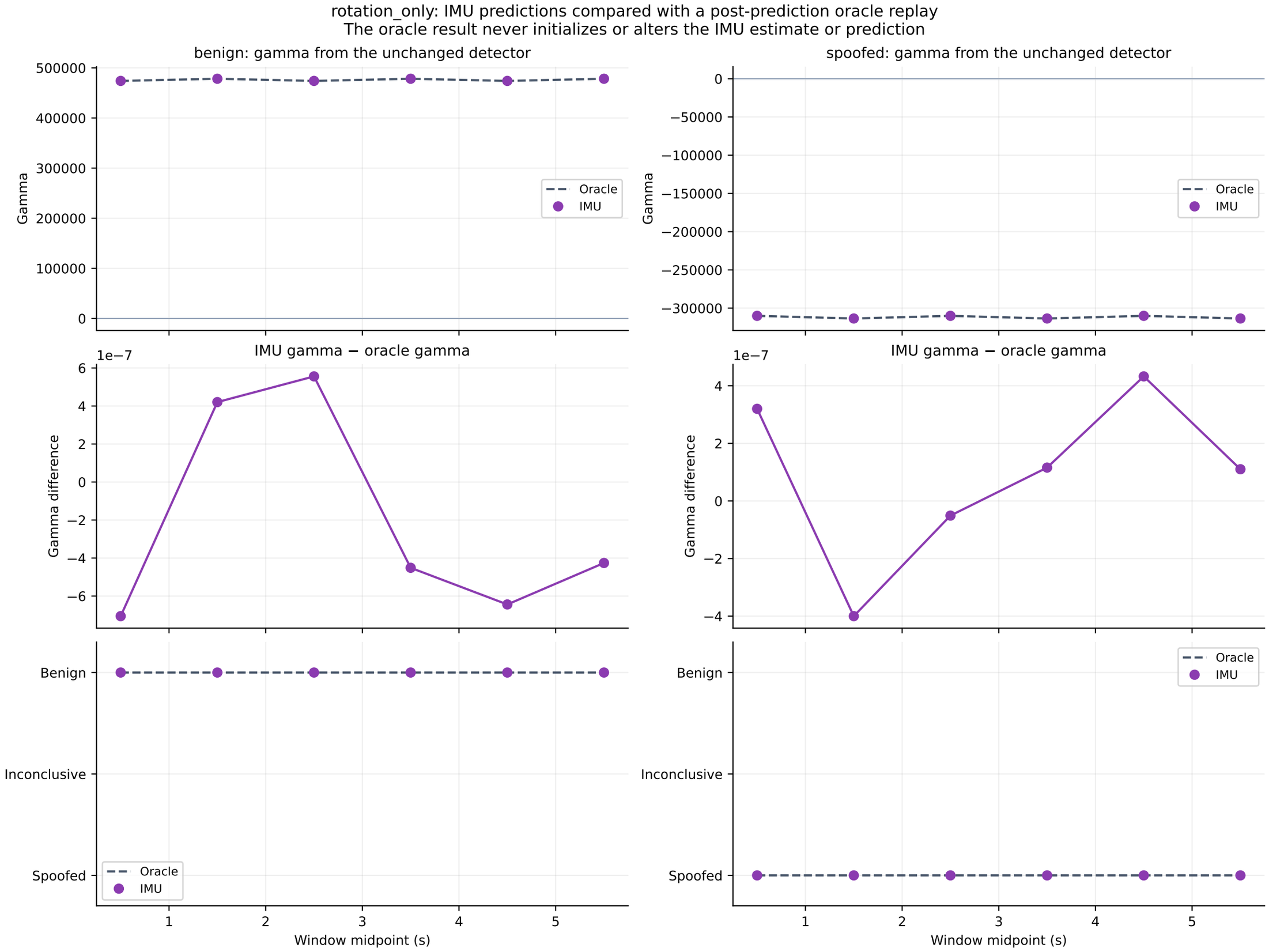

three_axis/benign: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
1e−5
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
Whitened fitted residual distribution
Density
Whitened residual

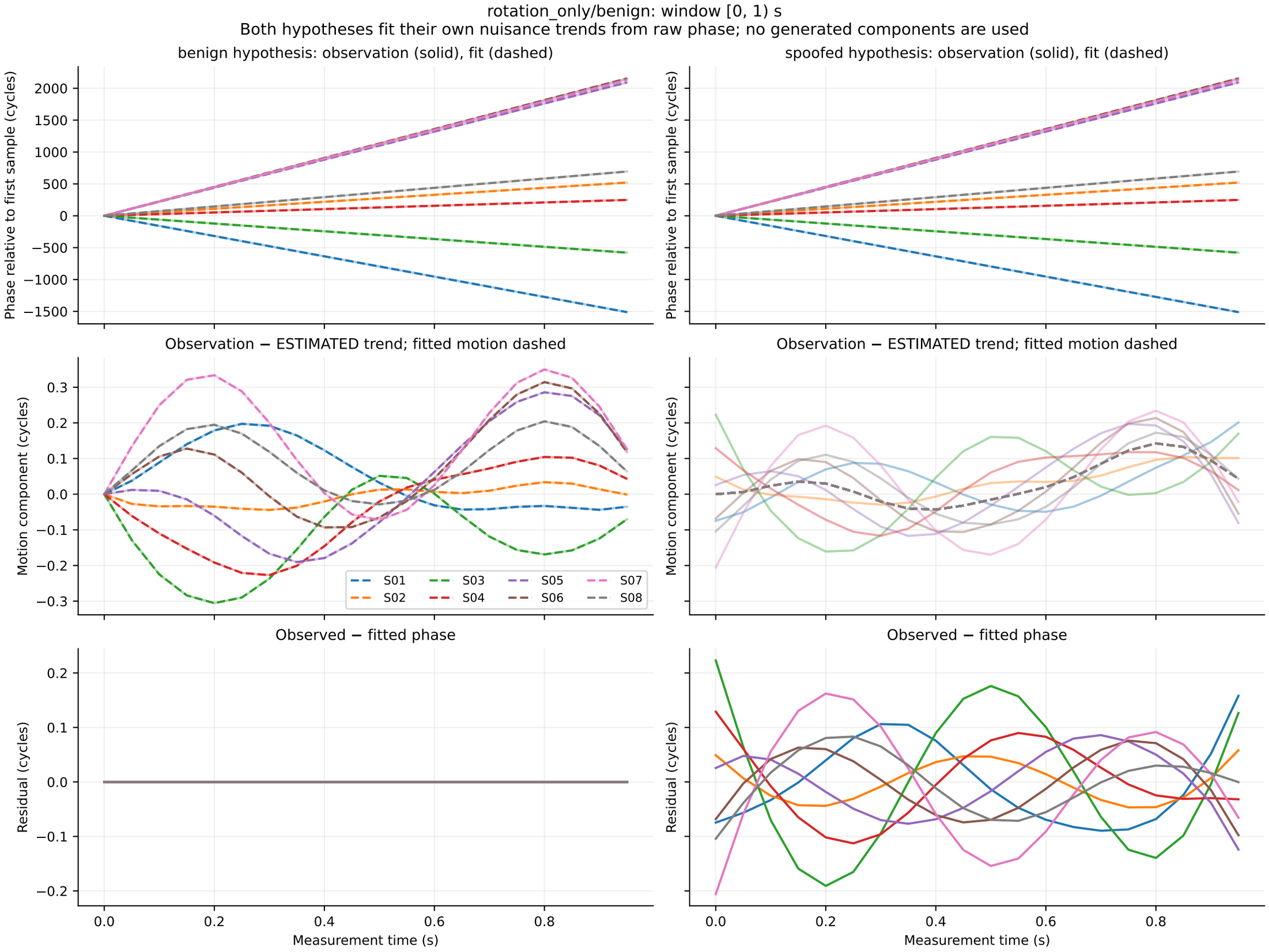

three_axis/spoofed: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
benign hypothesis: observation (solid), fit (dashed)
spoofed hypothesis: observation (solid), fit (dashed)
Phase relative to first sample (cycles)
Observation − ESTIMATED trend; fitted motion dashed
Motion component (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)

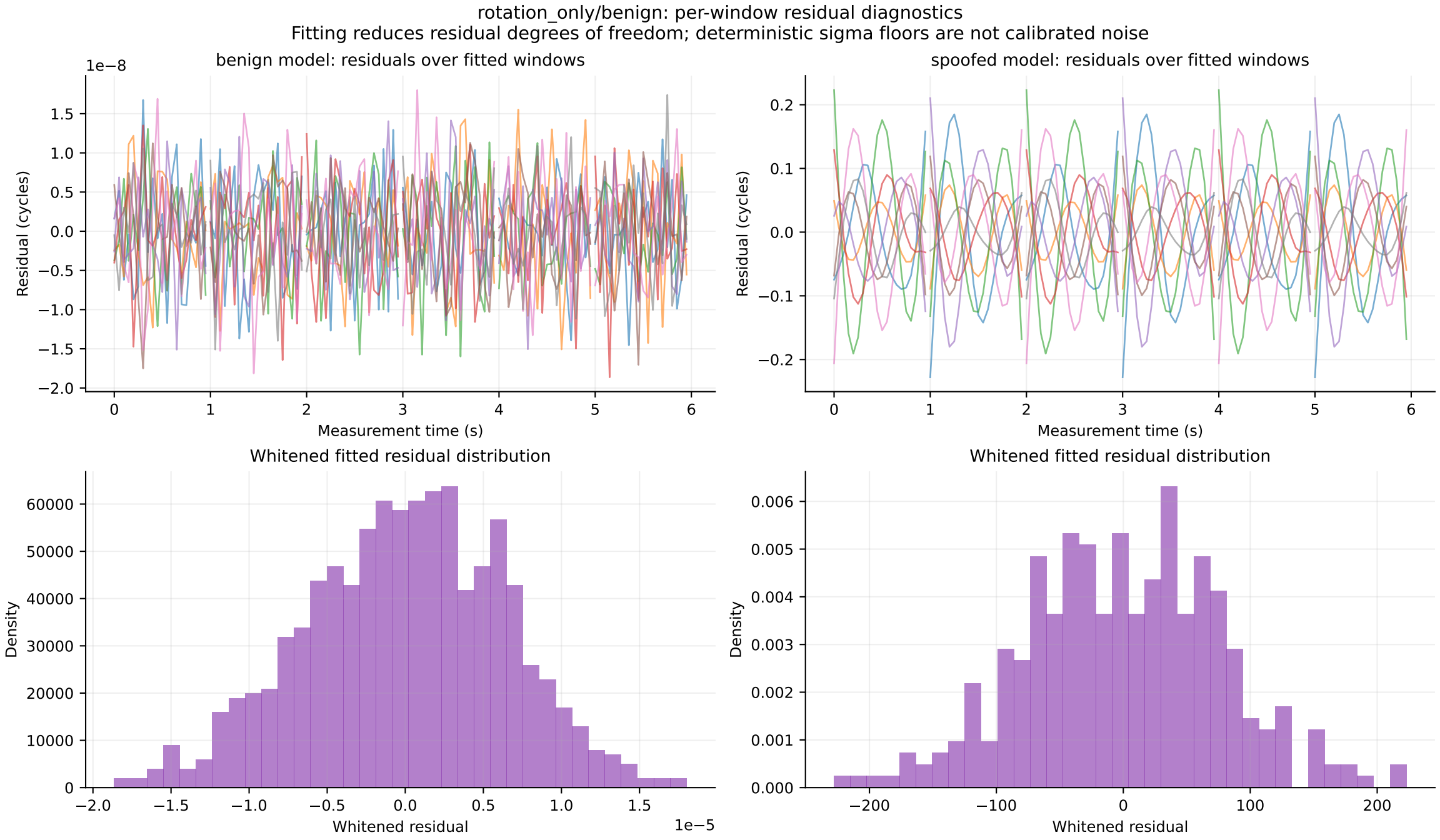

three_axis/spoofed: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
1e−5
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
Whitened fitted residual distribution
Density
Whitened residual

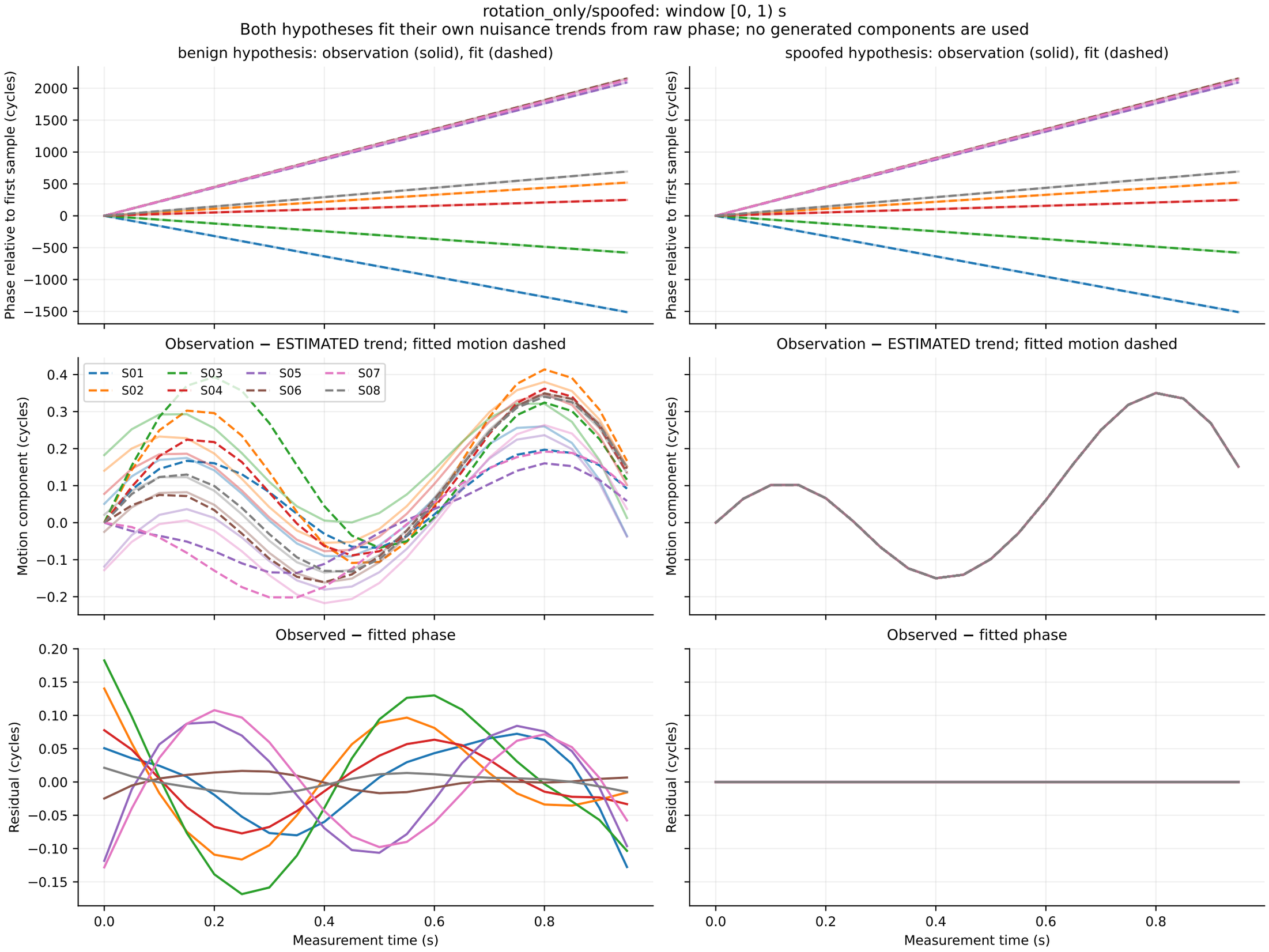

rotation_only: imu-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
Reduced negative log-likelihood
benign
spoofed
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

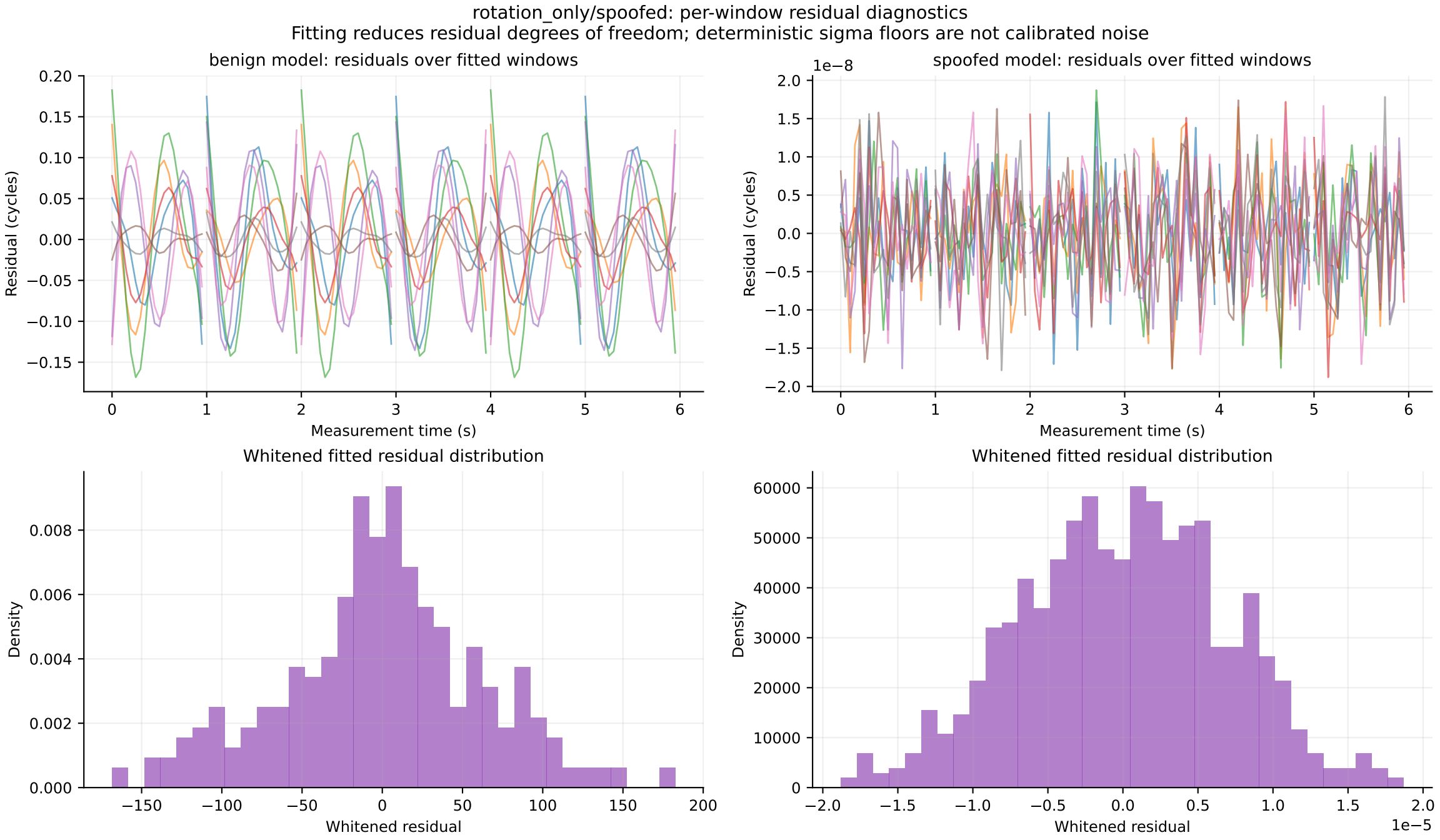

rotation_only: window [0, 1) s — observable motion RMSE 3.98e-11 µm
Solid: held-out truth; dashed: zero-velocity-initialized IMU estimate. Truth was read after prediction.
Raw: unknown initial velocity gives linear drift
East truth
North truth
Up truth
Antenna displacement ENU (m)
Measurement time (s)
After quadratic removal: motion used by detector
benign: true and estimated observable motion phase
S01
S02
S03
S04
S05
S06
S07
S08
Phase (cycles)
Estimated − true phase after polynomial removal
1e−13
Phase error (millicycles)
Measured acceleration rotated to ENU, gravity removed
1e−15
Linear acceleration (m/s²)
East
North
Up
IMU time (s)
Antenna displacement due to rotation of the body offset
Lever displacement ENU (m)

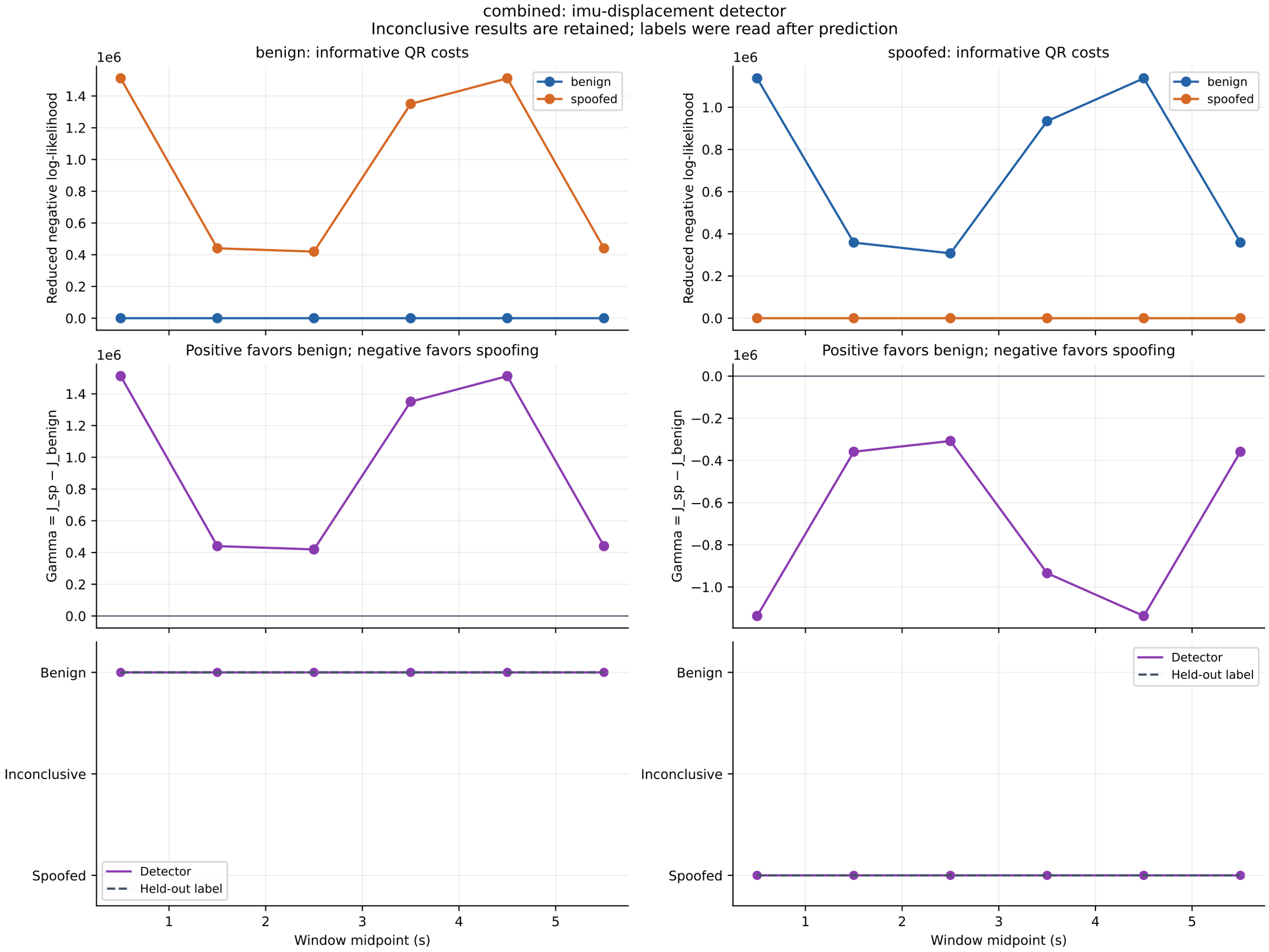

rotation_only: IMU predictions compared with a post-prediction oracle replay
The oracle result never initializes or alters the IMU estimate or prediction
benign: gamma from the unchanged detector
spoofed: gamma from the unchanged detector
Gamma
Oracle
IMU
IMU gamma − oracle gamma
1e−7
Gamma difference
Benign
Inconclusive
Spoofed
Window midpoint (s)

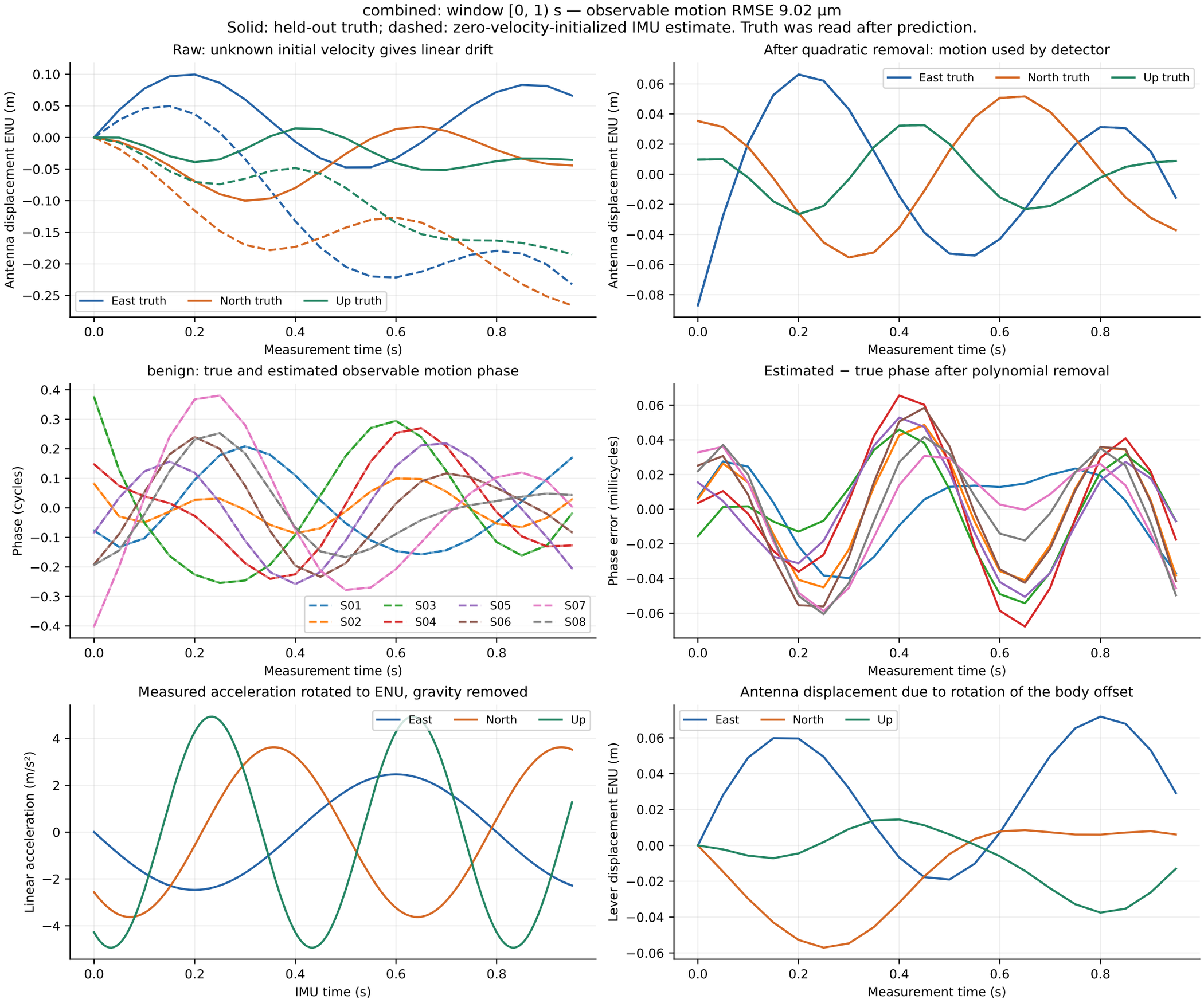

rotation_only/benign: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
benign hypothesis: observation (solid), fit (dashed)
spoofed hypothesis: observation (solid), fit (dashed)
Phase relative to first sample (cycles)
Observation − ESTIMATED trend; fitted motion dashed
Motion component (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)

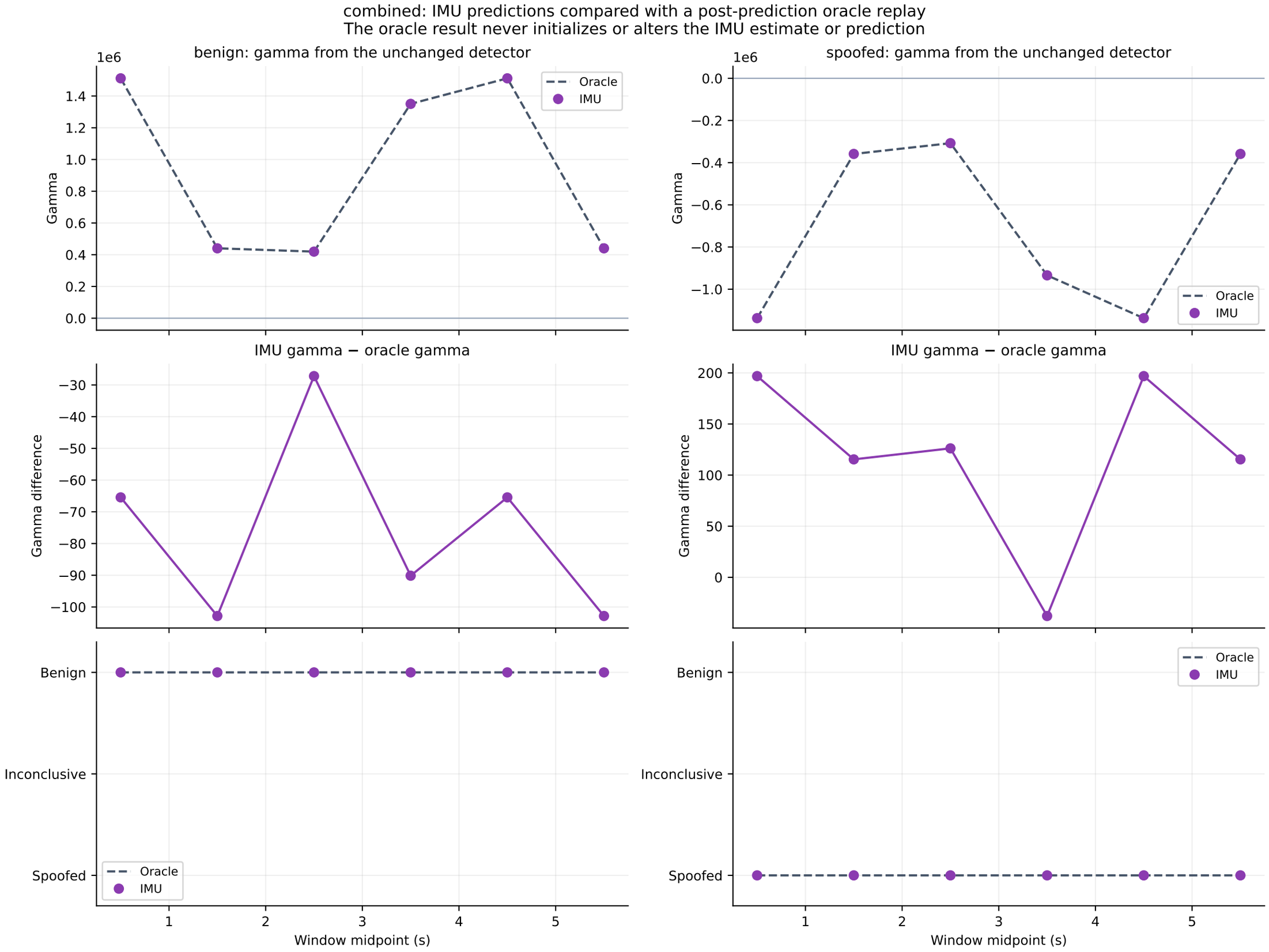

rotation_only/benign: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
1e−8
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
1e−5
Whitened fitted residual distribution
Density
Whitened residual

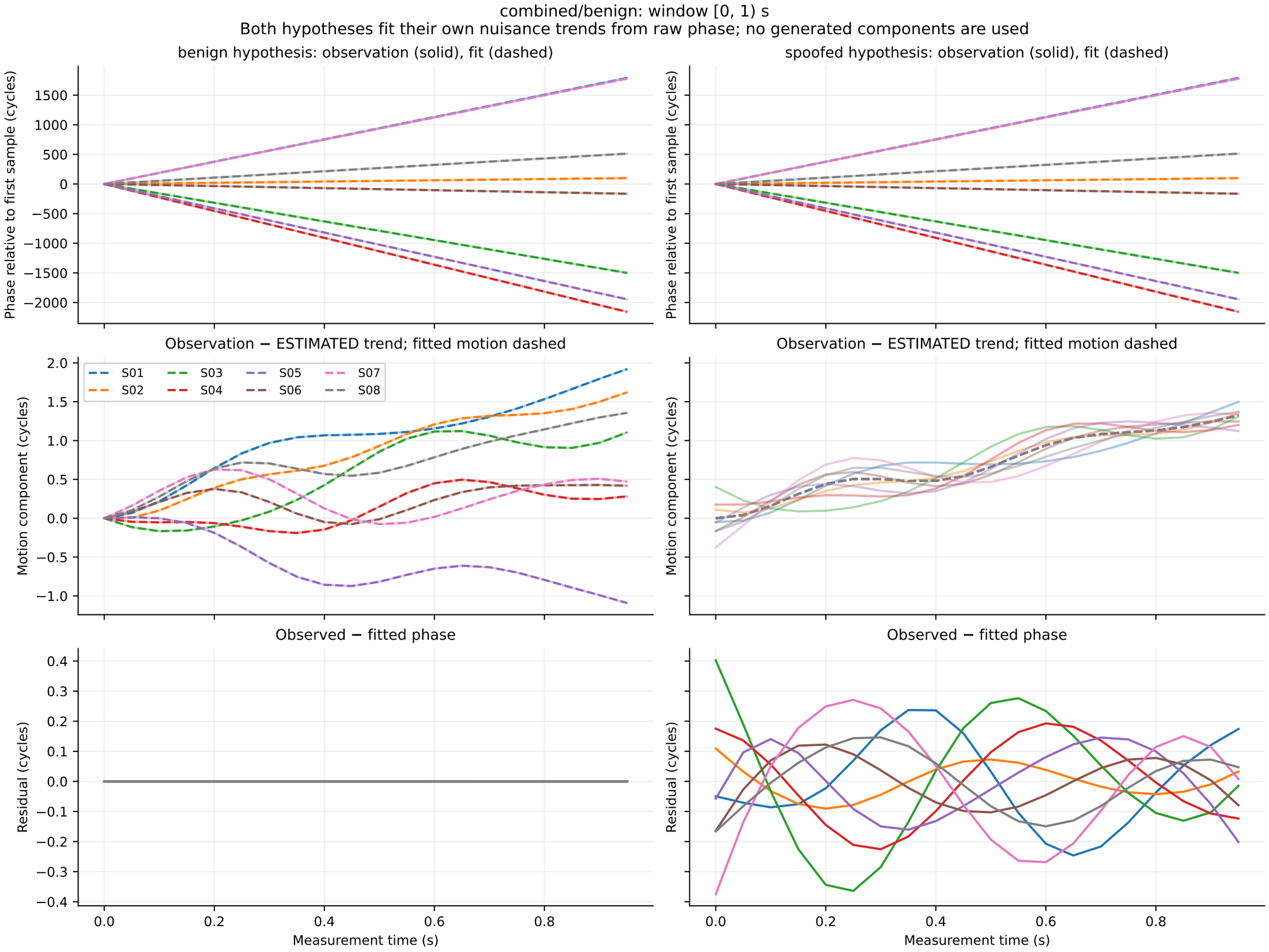

rotation_only/spoofed: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
benign hypothesis: observation (solid), fit (dashed)
spoofed hypothesis: observation (solid), fit (dashed)
Phase relative to first sample (cycles)
Observation − ESTIMATED trend; fitted motion dashed
Motion component (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)

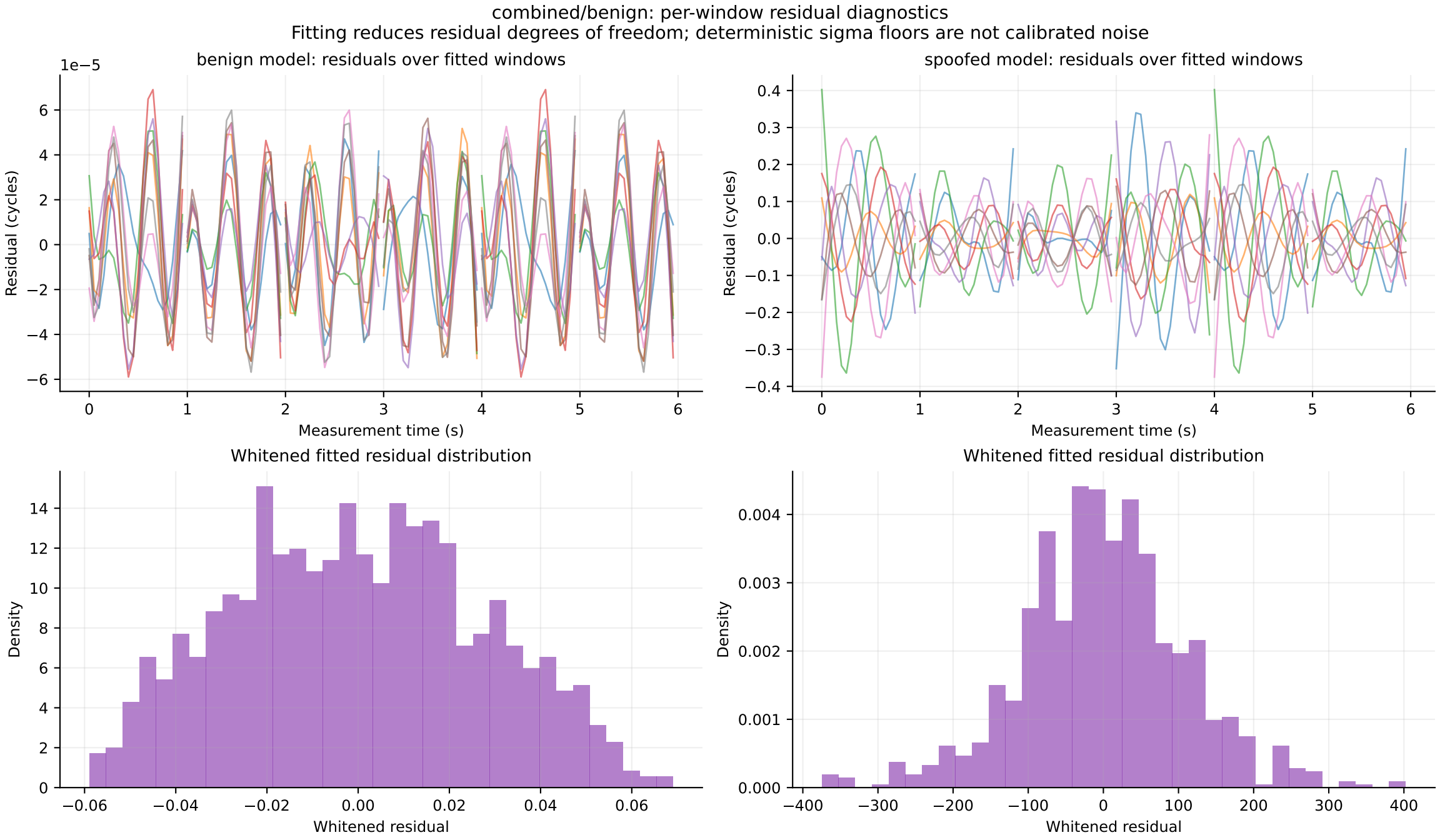

rotation_only/spoofed: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
1e−8
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
Whitened fitted residual distribution
Density
Whitened residual
1e−5

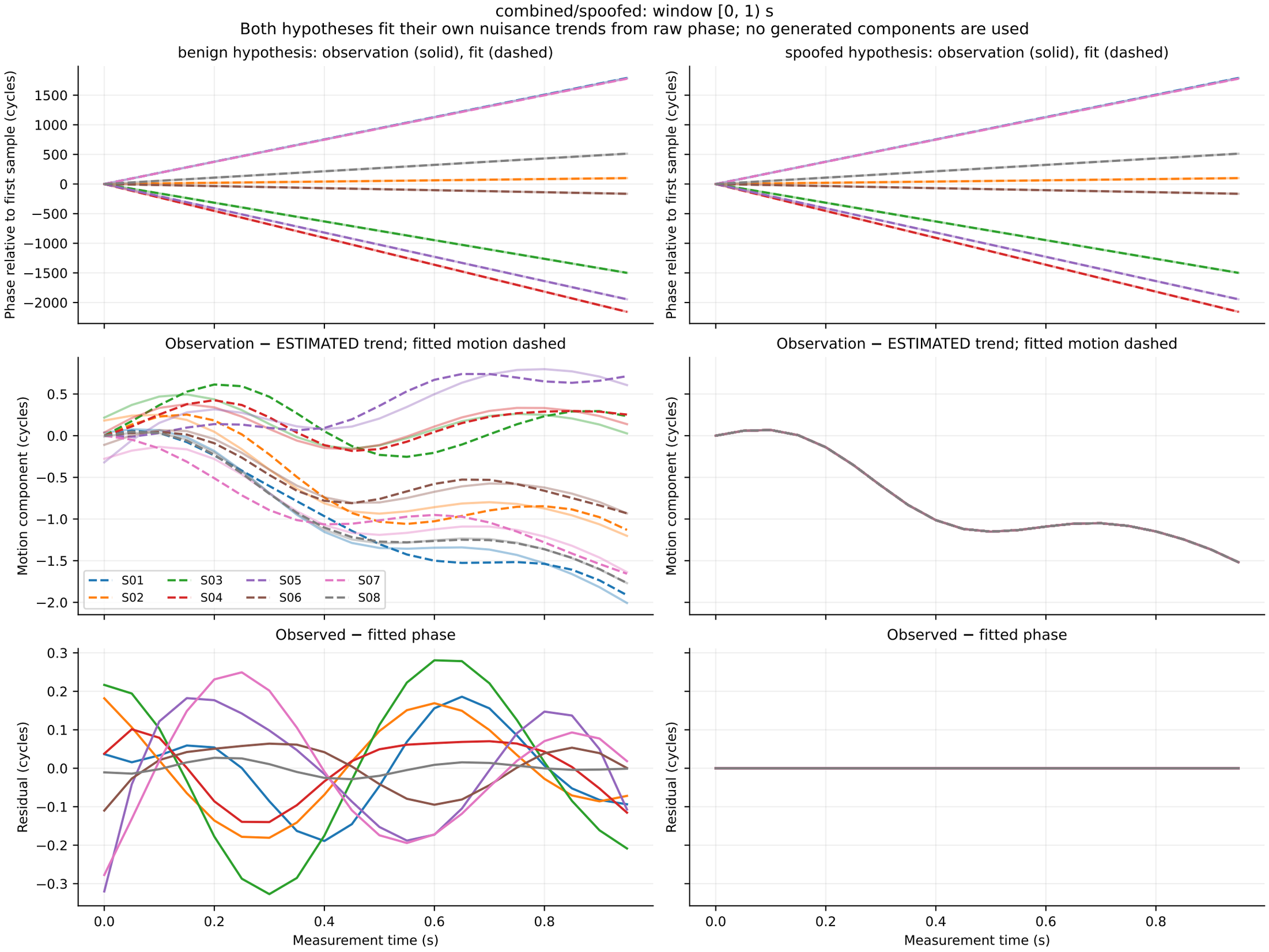

combined: imu-displacement detector
Inconclusive results are retained; labels were read after prediction
benign: informative QR costs
spoofed: informative QR costs
1e6
Reduced negative log-likelihood
benign
spoofed
Positive favors benign; negative favors spoofing
Gamma = J_sp − J_benign
Benign
Inconclusive
Spoofed
Detector
Held-out label
Window midpoint (s)

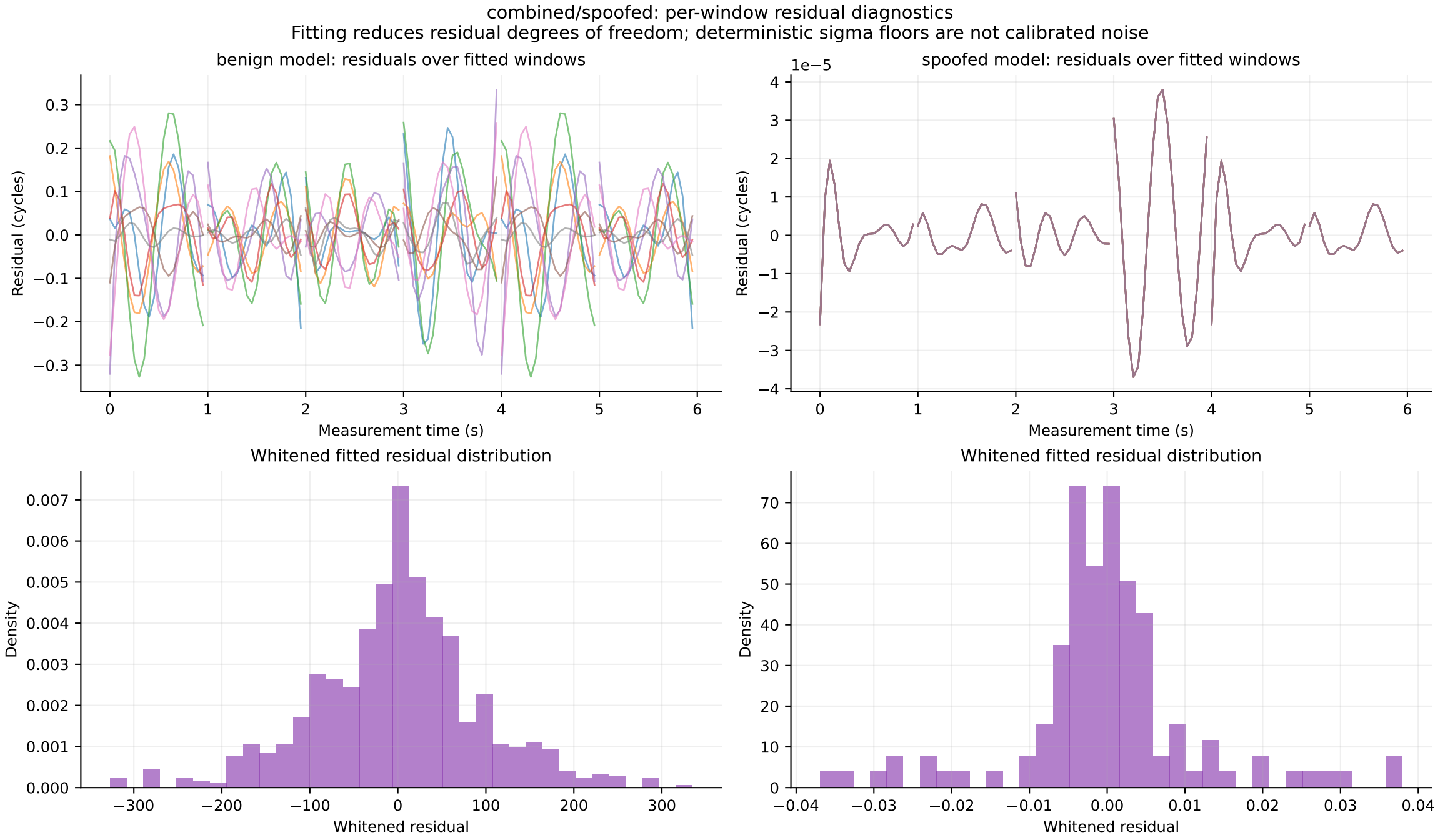

combined: window [0, 1) s — observable motion RMSE 9.02 µm
Solid: held-out truth; dashed: zero-velocity-initialized IMU estimate. Truth was read after prediction.
Raw: unknown initial velocity gives linear drift
Antenna displacement ENU (m)
East truth
North truth
Up truth
Measurement time (s)
After quadratic removal: motion used by detector
benign: true and estimated observable motion phase
Phase (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Estimated − true phase after polynomial removal
Phase error (millicycles)
Measured acceleration rotated to ENU, gravity removed
Linear acceleration (m/s²)
East
North
Up
IMU time (s)
Antenna displacement due to rotation of the body offset
Lever displacement ENU (m)


combined: IMU predictions compared with a post-prediction oracle replay
The oracle result never initializes or alters the IMU estimate or prediction
benign: gamma from the unchanged detector
spoofed: gamma from the unchanged detector
1e6
Gamma
Oracle
IMU
IMU gamma − oracle gamma
Gamma difference
Benign
Inconclusive
Spoofed
Window midpoint (s)


combined/benign: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
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Phase relative to first sample (cycles)
Observation − ESTIMATED trend; fitted motion dashed
Motion component (cycles)
S01
S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)


combined/benign: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
1e−5
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
Whitened fitted residual distribution
Density
Whitened residual


combined/spoofed: window [0, 1) s
Both hypotheses fit their own nuisance trends from raw phase; no generated components are used
benign hypothesis: observation (solid), fit (dashed)
spoofed hypothesis: observation (solid), fit (dashed)
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S02
S03
S04
S05
S06
S07
S08
Observed − fitted phase
Residual (cycles)
Measurement time (s)


combined/spoofed: per-window residual diagnostics
Fitting reduces residual degrees of freedom; deterministic sigma floors are not calibrated noise
benign model: residuals over fitted windows
Residual (cycles)
Measurement time (s)
spoofed model: residuals over fitted windows
1e−5
Residual (cycles)
Measurement time (s)
Whitened fitted residual distribution
Density
Whitened residual
Whitened fitted residual distribution
Density
Whitened residual